

## Full length article

## A quantum-linguistic multi-attribute group consensus method based on trust networks

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## ABSTRACT

In the context of uncertain social networks, belief biases and behavioral interactions among decision-makers often lead to interference effects during the decision-making process. To address this issue, this paper proposes a multi-attribute group consensus decision-making model based on quantum decision theory. Firstly, linguistic Z-numbers are extended into quantum states to represent uncertain information. The interference effects among trust paths are comprehensively considered, and trust relationships are aggregated based on quantum probability theory to complete the trust network. On this basis, the comprehensive relative out-degree centrality index is used to determine the weights of decision-makers. Secondly, a minimum adjustment group consensus optimization model based on trust relationships is constructed to analyze the impact of decision-makers' trust levels on group opinion adjustments. Furthermore, considering the risk preference psychology of decision-makers in the decision-making process, attributes' weights are determined using different probability distributions, and the score values of alternatives are calculated to obtain the ranking results. Finally, the proposed model is validated through a case study of production line configuration in an electronics manufacturing company, and sensitivity analysis and comparative analysis are conducted to demonstrate the rationality and robustness of the proposed method. The results show that the proposed model can effectively characterize interference effects and promote the achievement of group consensus.

## 1. Introduction

In modern society, the need for group decision-making (GDM) has increased significantly due to the complexity of information and the diversity of stakeholders [1,2]. This trend underscores the importance of achieving group consensus among diverse opinions [3], as consensus effectively coordinates differing views, promotes collective action, and optimizes decision outcomes. Furthermore, due to the complexities of information dissemination and the growing interdependence among decision-makers, multi-attribute group decision-making (MAGDM) methods have become increasingly critical in this context [4,5]. However, the limitations of knowledge and subjective biases of decision-makers can deteriorate decision quality, particularly in complex and dynamic environments. Additionally, in practical decision-making processes, it is often challenging to obtain accurate quantitative data, and some information cannot be easily represented by specific numbers. Consequently, linguistic information has gained attention as a vital tool for managing uncertainty and expressing fuzziness [6].

With the deepening of research, the forms of linguistic information representation have continuously evolved, including binary linguistic

numbers, hesitant fuzzy linguistic term sets, and probabilistic linguistic term sets [7–9]. These methods have enhanced the capacity to express uncertain information to some extent. However, in the face of complex decision-making scenarios, these traditional linguistic representations are often difficult to consider both confidence and fuzziness, and cannot fully express the limitations of decision makers' subjective uncertainty. To address these challenges, the linguistic Z-numbers (LZNs) has been introduced as a novel approach to information representation [10–12]. By integrating fuzzy numbers with probability theory, LZNs offer a more flexible means of expressing a decision maker's judgment under uncertain conditions. This approach not only accounts for the fuzziness inherent in linguistic information but also enhances the comprehensiveness and accuracy of information expression by providing a quantified range of credibility. Nevertheless, the performance of LZNs remains limited in more complex trust networks and when multiple interference effects are present. For instance, in the case of linguistic set {*bad*, *fair*, *good*}, the decision maker's evaluation may reflect an uncertain state between “fair” and “good”.

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With the rapid development of social networks, achieving group consensus in the multi-attribute decision-making process has emerged as a key research issue [13]. The complexity of this problem lies in the diversification of information transmission paths and the increasing mutual influence among decision makers, making it particularly important to effectively aggregate trust among decision makers under uncertain circumstances [14]. Traditional group consensus decision models typically rely on linear or independent trust paths to complete the trust network, assuming no mutual interference between paths [15]. However, in actual social networks, trust paths between decision makers often influence each other, and this interference effect is challenging for traditional decision models to capture, leading to insufficient accuracy in trust network completion. Additionally, traditional group consensus models typically depend on consensus measurement and opinion adjustment methods to gradually converge members' preferences [16]. For instance, Ji et al. [17] proposed a decay trust propagation method and a trust-driven consensus model for group decision-making in social networks, emphasizing the use of overlapping nodes and trust decay to improve consensus efficiency and realism. Feng et al. [18] introduced a semi-supervised learning model to incorporate unlabeled decision makers into social network group decision-making, enhancing classification accuracy and reducing consensus costs. Ma et al. [19] proposed a maximum expert consensus model, which integrates social relations and uncertain adjustment costs, providing a robust solution to address cost uncertainty in decision-making. Wu et al. [20] introduced a consensus model into social networks, utilizing linguistic trust propagation and a maximum self-esteem feedback mechanism to balance individual self-esteem with group consensus. However, these methods still face limitations in practical application, particularly the insufficient consideration of mutual trust levels in the adjustment of decision makers' preferences. In actual decision-making environments, preference adjustment is not only constrained by the degree of confidence but may also be influenced by the level of trust in one's own evaluation. If the adjustment extends beyond the confidence range, it may incur additional costs and negatively impact the quality of the group consensus.

In the process of multi-attribute decision-making, the determination of attribute weights is a critical factor influencing decision quality. Traditional methods, such as the Analytic Hierarchy Process (AHP), the entropy weight method, and TODIM [21–23], typically rely on expert judgment or objective indicators derived from data to establish attribute weights. While these methods can effectively reflect the importance of individual attributes to some extent, they often assume that the decision environment is deterministic and that decision makers' preferences remain stable. In practice, however, the psychological state of decision makers, particularly their risk preferences, such as risk aversion or risk seeking [24], can significantly impact the allocation of attribute's weights. Moreover, in uncertain decision-making environments, the distribution of attribute weights may vary as the environment changes. Traditional methods, however, tend to inadequately account for these dynamic changes, often overlooking the variability of weights in different situations.

Quantum probability theory (QPT), as an emerging probability framework, has demonstrated distinct advantages in multi-attribute decision-making compared to classical probability theory [25,26]. Traditional multi-attribute decision-making methods rely on classical probability theory, which assumes that decision information is independent and additive. However, in complex decision-making environments, interactions often occur between decision makers' preferences and attributes, which classical probability theory struggles to accurately model [27,28]. Quantum probability theory leverages quantum superposition and interference phenomena to simultaneously consider multiple decision preferences, optimizing the decision process through these interactions [29]. In QPT, interference effects refer to the interactions between different decision preferences that can alter otherwise independent decision outcomes, thereby affecting the final probability

distribution. To achieve this, the final quantum probability value can be calculated by constructing a quantum-like Bayesian network (QLBN) and using the interference angle to represent the intensity of mutual interference between decision makers [30,31]. This property makes QPT more flexible and accurate in addressing multi-attribute decision-making problems characterized by high uncertainty and complex interference, providing new theoretical tools and methods for complex decision-making [32,33]. For instance, Yan et al. [34] proposed an interference effect model using minimum constraints and information entropy to address the quantum entanglement problem in multi-attribute group decision-making. Tan et al. [35] introduced a consensus improvement mechanism that combines prospect theory with quantum probability theory to address irrational behavior in linguistic group decision-making. Wang et al. [36] developed a two-mechanism model that integrates trust exploration and leadership incubation to effectively explore the evolution of opinions in social networks during the process of reaching consensus on group decisions.

Although significant progress has been made in the current research, several key issues remain unresolved in the development of multi-attribute group consensus decision theory. Specifically, these issues focus on the following three areas:

(1) The use of quantum linguistic Z-numbers (QLZN) in MAGDM has not been fully explored. While the potential of quantum decision theory in addressing complex decision problems is increasingly recognized, the specific application of QLZN remains relatively limited. This is particularly evident in their ability to manage complex interactions and information uncertainties among decision makers in group decision-making contexts.

(2) Existing consensus models for social networks exhibit certain limitations in handling trust relationships between decision makers and mutual interference within trust paths. These limitations may undermine the accuracy and stability of the consensus process in complex network environments. Therefore, developing a group consensus model that better integrates trust relationships and accounts for trust path interference could more accurately reflect real-world decision-making environments and facilitate more effective group consensus.

(3) The determination of the weights of attributes often overlooks the decision maker's mental state and environmental uncertainty. Current studies frequently neglect the impact of decision makers' psychological factors and external uncertainties on weight allocation. Thus, exploring how to comprehensively incorporate decision makers' psychological factors and environmental uncertainties in uncertain environments to determine more reasonable weights of attributes remains a pressing need.

A comparison between this model and existing decision models is presented in Table 1.

As shown in Table 1, existing traditional group consensus models often focus on the consensus-reaching process itself, emphasizing the influence of consensus mechanisms on achieving group opinion consistency. However, these models show significant shortcomings in the objective calculation of decision-makers' weights and attributes' weights, which are often assigned subjectively, lacking objectivity and scientific basis. Moreover, they seldom consider the impact of trust relationships among decision-makers on the consensus results. In addition, some models fail to effectively address the complexity and uncertainty of the decision-making environment, which limits their practical applicability. Although current multi-attribute quantum group decision-making models introduce quantum interference effects and enhance the ability to describe nonlinear interactions in the decision-making process, most of these studies still rely on real numbers to represent decision-makers' information. This approach does not sufficiently reflect the uncertainty of precise data or the difficulty in obtaining complete information. Furthermore, these studies tend to oversimplify the calculation of decision-makers' and attributes' weights and pay little attention to the dynamic adjustment mechanisms required for the group consensus process.

**Table 1**

The comparison of the proposed method and previous decision models..

| Literature         | Information | DMS' weights | Group consensus | Trust relationship | Attributes' weights | Interference effect |
|--------------------|-------------|--------------|-----------------|--------------------|---------------------|---------------------|
| [37]               | CN          | Known        | Yes             | No                 | Known               | No                  |
| [38]               | LTS         | Unknown      | Yes             | Yes                | No                  | No                  |
| [39]               | IFS         | Unknown      | Yes             | Yes                | No                  | No                  |
| [40]               | CN          | Known        | Yes             | No                 | Known               | No                  |
| [41]               | CN          | Known        | Yes             | No                 | Known               | No                  |
| [31]               | CN          | No           | No              | Yes                | Yes                 | Yes                 |
| [33]               | LTS         | Unknown      | No              | Unknown            | Unknown             | Yes                 |
| [42]               | CN          | Unknown      | No              | No                 | Unknown             | Yes                 |
| [36]               | CN          | Unknown      | No              | No                 | Known               | Yes                 |
| [30]               | LTS         | Known        | Yes             | No                 | No                  | Yes                 |
| The proposed model | QLZN        | Unknown      | Yes             | Yes                | Unknown             | Yes                 |

Note: CN is Crisp number, LTS is Linguistic term set, IFS is Intuitionistic fuzzy set.

In contrast, this paper proposes a quantum linguistic Z-numbers based multi-attribute group consensus model in social networks that incorporates quantum interference effects. The model innovatively employs quantum linguistic Z-numbers to represent decision information, which not only better captures uncertainty and fuzziness but also introduces phase information to reflect potential interference effects among decision-makers' evaluations. Furthermore, trust relationships in incomplete trust networks are aggregated based on quantum probability theory, and decision-makers' weights are determined objectively using the comprehensive relative out-degree centrality index. In terms of consensus optimization, a minimum adjustment group consensus optimization model based on trust relationships is constructed to dynamically adjust group opinions and achieve a high level of consensus. In addition, this study fully considers the risk preference psychology of decision-makers by determining attribute weights through different probability distributions combined with Monte Carlo simulation.

The main contributions of this paper are as follows:

(1) The LZN is extended to a quantum state form and applied to a multi-attribute group decision model to accurately describe and quantify the complex interactions and information uncertainties among decision makers. This innovation not only expands the application domain of QLZN but also significantly enhances the accuracy and flexibility of decision models in handling complex decision scenarios.

(2) The existing group consensus model is advanced by incorporating trust relationships and path interference effects. Path interference is quantified to complete the trust network, and the influence of trust relationships on the group consensus process is examined, thereby more accurately reflecting the complex trust dynamics between decision makers. This approach improves the accuracy and stability of achieving group consensus in real-world social networks.

(3) A maximum utility model based on risk preference is introduced to calculate attribute weights, with the attribute weight distribution dynamically adjusted by simulating decision makers' risk preferences in various situations. This method not only more accurately reflects the psychological states of decision makers but also optimizes weight allocation in uncertain environments, improving the accuracy and flexibility of decision-making.

The paper is organized as follows: Section 2 introduces the LZN, trust network and related concepts of quantum probability theory. In Section 3, the QLZNs multi-attribute quantum group consensus model based on trust network is introduced, and the decision steps are introduced in detail. In Section 4, the model is verified by a production line configuration example of an electronics company, and the stability and reliability of the model are verified by sensitivity analysis and comparative analysis. Finally, Section 5 summarizes the main findings and contributions of this paper, and suggests future research directions.

## 2. Preliminaries

### 2.1. Linguistic term set

Let  $L = \{l_t | t = 0, 1, \dots, 2g\}$  be a finite and ordered set of linguistic terms with an odd number of terms, where  $g$  is a non-negative integer, ensuring evaluation symmetry and a clear neutral option to accurately reflect decision-makers' true preferences.

**Definition 1** ([43]). If  $\theta_t \in [0, 1]$  is a real number, then the linguistic scale function  $F$  represents the mapping from  $l_t$  to  $\theta_t$  with the following expression:

$$F : l_t \rightarrow \theta = \theta_t, t = 1, 2, \dots, 2g. \quad (1)$$

### 2.2. Linguistic Z-numbers

The linguistic Z-numbers is a linguistic evaluation method based on fuzzy theory, which is composed of two linguistic variables: the first one is used to describe the fuzziness of the evaluation results, and the second one is used to represent the credibility of the evaluation content. Fuzzy theory provides mathematical tools for linguistic Z-number, which enables it to effectively express the fuzziness and reliability of evaluation under uncertainty conditions, and is suitable for information representation in complex decision scenarios.

**Definition 2** ([28]). Suppose  $X$  is a domain of discourse,  $L = \{l_0, l_1, \dots, l_{2g}\}$  and  $L' = \{l'_0, l'_1, \dots, l'_{2g'}\}$  are two sets with an odd number of finite, completely ordered discrete linguistic terms. If  $A_{\varphi(x)} \in L$  and  $B_{\phi(x)} \in L'$ , then a LZN of set  $X$  is defined as follows:

$$Z = \{(x, A_{\varphi(x)}, B_{\phi(x)}) | x \in X\}. \quad (2)$$

where  $A_{\varphi(x)}$  represents the fuzzy constraint on event  $X$ , and  $B_{\phi(x)}$  is the reliability measure for the first part. When  $X$  has only one element, the LZN is changed to  $(A_{\varphi(x)}, B_{\phi(x)})$ .

### 2.3. Social network

**Definition 3** ([36]). A social network is a directed graph  $G(D, E)$ , where  $D = \{d_1, d_2, \dots, d_s\} (s \geq 3)$  represents the set of experts,  $E = \{e_1, e_2, \dots, e_q\}$  is the set of directed lines or arcs between nodes, side  $(e_k, e_h) \in E$  represents the expert  $d_k$  directly affecting the expert  $d_h (k, h \in s)$ , and a sociometric  $T = (t_{kh})_{s \times s} (t_{kh} \in \{0, 1\})$  is used to represent  $G(D, E)$ :

$$t_{kh} = \begin{cases} 1, & (d_k, d_h) \in E \\ 0, & \text{other} \end{cases} \quad (3)$$

$$k = 1, 2, \dots, s, h = 1, 2, \dots, s, k \neq h.$$

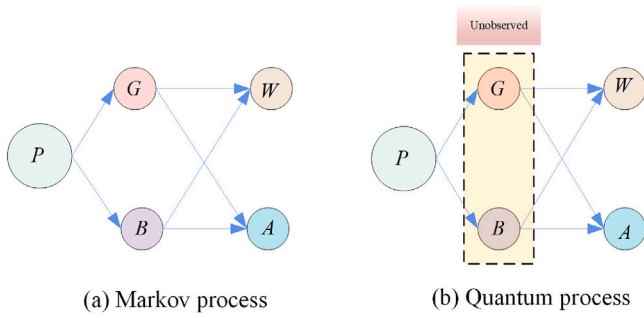


Fig. 1. Markov chain path diagram.

## 2.4. Quantum probability theory

The application of quantum probability theory to decision theory is a relatively new research field [25]. Unlike traditional decision theory, which assumes that individuals evaluate all possibilities when faced with various options and make optimal decisions accordingly, quantum probability theory introduces concepts from quantum mechanics, such as superposition states and quantum entanglement, to explain some of the non-classical behaviors observed in human decision-making, as illustrated in Fig. 1. Fig. 1 illustrates the difference between a classical Markov process and a quantum process in decision-making. In the quantum process (Fig. 1(b)), an “unobserved” state is introduced, capturing quantum superposition and entanglement effects, which help explain non-classical behaviors in human decision-making.

In quantum Bayesian networks, probability amplitude is used to represent probability, and the magnitude of the probability is equal to the square of the projected length [27]. According to Feynman’s first rule, the probability amplitude of a path is the product of the probability amplitudes along that path. Thus, the probability amplitude of choosing  $W$  along path  $P \rightarrow G \rightarrow W$  is  $\psi_G \psi_{W|G}$ , and the probability amplitude of choosing  $A$  along path  $P \rightarrow B \rightarrow A$  is  $\psi_B \psi_{B|A}$ . In real-life scenarios, the specific path followed by a decision cannot be clearly observed, and the system remains in a superposition state along the path. According to Feynman’s second rule, if the specific path of the state transition is not observed, the probability of the final transition is calculated by first summing the probability amplitudes of all possible paths and then taking the square of that sum.

$$\begin{aligned}
 P(W) &= |\psi_G \psi_{W|G} + \psi_B \psi_{B|A}|^2 \\
 &= |\varphi_G e^{i\theta_G} \varphi_{A|G} e^{i\theta_{A|G}} + \varphi_B e^{i\theta_B} \varphi_{B|A} e^{i\theta_{B|A}}|^2 \\
 &= |\varphi_G e^{i\theta_G} \varphi_{A|G} e^{i\theta_{A|G}} + \varphi_B e^{i\theta_B} \varphi_{B|A} e^{i\theta_{B|A}}|^2 \\
 &= P(G) P(W|G) + P(B) P(B|A) \\
 &\quad + 2\sqrt{P(G) P(W|G) P(B) P(B|A)} \cos(\theta_G - \theta_B).
 \end{aligned} \quad (4)$$

## 3. Quantum linguistic Z-number multi-attribute group consensus model considering quantum interference in trust networks

The main goal of a MAGDM is to derive a consensus-based solution from multiple alternatives with multiple attributes. The typical elements of multi-attribute group decision problems are as follows.

(1) There are  $s$  decision makers  $D = \{d_1, d_2, \dots, d_s\}$  ( $s \geq 3$ ), the weight set of decision makers is  $\lambda = \{\lambda_1, \lambda_2, \dots, \lambda_s\}$ , and the conditions  $\lambda_k \geq 0$  ( $k = 1, 2, \dots, s$ ) and  $\sum_{k=1}^s \lambda_k = 1$  are satisfied.

(2) The limited set of alternative alternatives is  $A = \{a_1, a_2, \dots, a_m\}$  ( $m \geq 2$ ), which are potential solutions to the multi-attribute group decision problem.

(3) The attribute set is  $C = \{c_1, c_2, \dots, c_n\}$  ( $n \geq 2$ ), the attributes’ weights set is  $W = \{\omega_1, \omega_2, \dots, \omega_n\}$ , and conditions  $\omega_j \geq 0$  ( $j = 1, 2, \dots, n$ ) and  $\sum_{j=1}^n \omega_j = 1$  are satisfied.

## 3.1. Evaluation information representation based on quantum linguistic Z-numbers

Traditional linguistic Z-numbers have certain limitations when dealing with highly complex and uncertain systems, as they are prone to information loss, insufficient expressive capacity, and difficulty in capturing dynamic interference and nonlinear interactions among decision factors. However, extending linguistic Z-numbers into quantum states can leverage the superposition and interference effects of quantum states, introducing phase information to capture the potential associations and interferences among factors in complex decision-making scenarios, thereby significantly improving the flexibility of information representation. In addition, the use of tensor product forms allows for the natural combination of multiple quantum states, reflecting multidimensional interactions and system coupling characteristics, and thus better handling uncertainty, fuzziness, and complexity. Therefore, this paper extends linguistic Z-numbers into quantum states and uses quantum linguistic Z-numbers to represent the evaluation information of decision-makers, in order to enhance applicability and expressive capacity under uncertain social network environments.

### 3.1.1. The quantum state form extension of LZNs

Suppose that  $H_L$  and  $H_{L'}$  are Hilbert Spaces corresponding to linguistic sets  $L$  and  $L'$ , respectively.

**Definition 4.** Every linguistic term  $l_i, l'_i$  in sets  $L$  and  $L'$  can be expressed as an orthogonal basis  $|l_i\rangle, |l'_i\rangle$  in  $H_L$  and  $H_{L'}$ , then the quantum state form of LZN can be expressed as:

$$|Z\rangle = \{(x, e^{i\theta_{\varphi(x)}} |A_{\varphi(x)}\rangle \otimes e^{i\theta_{\phi(x)}} |B_{\phi(x)}\rangle) | x \in X \}. \quad (5)$$

where, the linguistic terms  $A_{\varphi(x)}$  and  $B_{\phi(x)}$  represent quantum states  $e^{i\theta_{\varphi(x)}} |A_{\varphi(x)}\rangle \in H_L$  and  $e^{i\theta_{\phi(x)}} |B_{\phi(x)}\rangle \in H_{L'}$  with phase information, respectively, and  $\theta_{\phi(x)}$  and  $\theta_{\varphi(x)}$  represent phase angles.

When  $X$  has only one element, the QLZN can be simplified to:

$$|Z\rangle = e^{i\theta_{\varphi(x)}} |A_{\varphi(x)}\rangle \otimes e^{i\theta_{\phi(x)}} |B_{\phi(x)}\rangle$$

**Example 1.** Suppose there are two sets of linguistic numbers  $L = \{bad, fair, good\}$  and  $L' = \{impossible, possible, affirmative\}$ , and the linguistic set term is represented as the orthogonal basis of Hilbert space, then the quantum form of the sets can be expressed as follows:

$$|L\rangle = \{e^{i\theta_0} |bad\rangle, e^{i\theta_1} |fair\rangle, e^{i\theta_2} |good\rangle\}$$

and

$$|L'\rangle = \{e^{i\theta'_0} |impossible\rangle, e^{i\theta'_1} |possible\rangle, e^{i\theta'_2} |affirmative\rangle\}$$

Then the quantum state of the LZN  $Z = (bad, possible)$  can be represented as:  $|Z\rangle = e^{i\theta_0} |bad\rangle \otimes e^{i\theta'_1} |possible\rangle$ .

**Definition 5.** In order to embody the superposition between linguistic terms, it is extended to the following form:

$$|\psi_x\rangle = \sum_i \alpha_i e^{i\theta_{\varphi_i(x)}} |A_{\varphi_i(x)}\rangle \otimes \sum_j \beta_j e^{i\theta_{\phi_j(x)}} |B_{\phi_j(x)}\rangle. \quad (6)$$

where  $\alpha_i$  and  $\beta_j$  are complex coefficients and meet the normalization conditions  $\sum_i |\alpha_i|^2 = 1$  and  $\sum_j |\beta_j|^2 = 1$ , then the QLZN can be expressed as:

$$|Z\rangle = \{(x, |\psi_x\rangle) | x \in X\}. \quad (7)$$

**Example 2.** Based on the linguistic set of Example 1, suppose decision maker  $d_1$  evaluates the quality of alternative  $a_1$  as a superposition

between “good” and “fair”, and evaluates the possibility as a superposition between “possible” and “affirmable”, then it is expressed as follows:

$$|\psi_{d_1|a_1}\rangle = \left( |\sqrt{0.7}e^{i\theta_{good}}|good\rangle + |\sqrt{0.3}e^{i\theta_{fair}}|fair\rangle \right) \otimes \left( |\sqrt{0.6}e^{i\theta_{possible}}|possible\rangle + |\sqrt{0.4}e^{i\theta_{affirmable}}|affirmable\rangle \right).$$

Thus, with QLZN, decision makers can combine quantified values and phase information to fully describe their evaluations. This representation method not only accounts for the fuzziness of linguistic terms but also incorporates quantum superposition and interference effects, making the evaluation information richer and more accurate. Consequently, it provides a more powerful tool for addressing complex decision problems.

### 3.1.2. Crisp number transformation of QLZN

When the QLZN is used as decision information, direct comparison and calculation are often not possible. Therefore, to make the decision information more scientific and effective and the decision process more convenient, a quantitative method is needed to convert linguistic decisions into corresponding specific values.

In this paper, the expected value method is employed to convert QLZN into crisp number. QLZN consist of several linguistic terms along with their corresponding quantized values and phase information. By selecting appropriate observables  $O$ , such as diagonal matrices or projection operators, the expected value method calculates the weighted sum of these quantized values in quantum states. Compared to other methods, such as von Neumann entropy [44] and relative entropy [45], the expected value method avoids the issue of results approaching zero and is capable of handling complex systems containing multiple quantum states. Where, this paper assumes that the observation matrix  $o$  is a diagonal matrix based on quantum state aggregation.

Firstly, the first linguistic set quantum state expectation value of QLZN is calculated as follows:

$$\langle O \rangle_\psi = \left( \sum_i \alpha_i e^{-i\theta_i} \langle A_{\varphi_i(x)} | \right) O \left( \sum_i \alpha_i e^{i\theta_i} | A_{\varphi_i(x)} \rangle \right). \quad (8)$$

Secondly, the quantum state expected value of the second linguistic set is calculated:

$$\langle O' \rangle_\phi = \left( \sum_j \beta_j e^{-i\phi_j} \langle B_{\phi_j(x)} | \right) O' \left( \sum_j \beta_j e^{i\phi_j} | B_{\phi_j(x)} \rangle \right). \quad (9)$$

Finally, the whole state of the composite system can be expressed as the tensor product of some states, so QLZN is transformed into a crisp number in product form.

$$|\hat{Z}\rangle = \langle O \rangle_\psi \langle O' \rangle_\phi. \quad (10)$$

### 3.1.3. Decision information collection

The evaluation information of each decision-maker  $d_k (k = 1, 2, \dots, s)$  on attributes  $c_j (j = 1, 2, \dots, n)$  of the alternative  $a_i (i = 1, 2, \dots, m)$  is represented by the QLZN  $z_{ij}^k = e^{i\theta_{\varphi(x)}} |A_{\varphi(x)}\rangle \otimes e^{i\theta_{\phi(x)}} |B_{\phi(x)}\rangle$ , where, the decision maker needs to provide two uncertain evaluation information to collect the LZNs data: one is the fuzzy evaluation of a certain attribute, and the other is the confidence of the fuzzy evaluation, and then the top decision maker collects and processes the QLZN, and the evaluation information matrix  $Z^k = (z_{ij}^k)_{m \times n} (k = 1, 2, \dots, s)$  is expressed as:

$$Z^k = \begin{pmatrix} z_{11}^k & z_{12}^k & \dots & z_{1n}^k \\ z_{21}^k & z_{22}^k & \dots & z_{2n}^k \\ \vdots & \vdots & \ddots & \vdots \\ z_{m1}^k & z_{m2}^k & \dots & z_{mn}^k \end{pmatrix}, k = 1, 2, \dots, s. \quad (11)$$

On this basis, the initial quantum state information evaluation matrix  $Z^k = (z_{ij}^k)_{m \times n} (k = 1, 2, \dots, s)$  can be transformed into a crisp number decision matrix  $\hat{Z}^k = (\hat{z}_{ij}^k)_{m \times n} (k = 1, 2, \dots, s)$  by the expected value method.

## 3.2. Group consensus model for trust networks considering quantum interference

### 3.2.1. Trust aggregation based on quantum interference

In a social trust network, nodes represent individual decision makers, and edges represent the degree of trust between them. However, in practical applications, trust networks are often incomplete, with many indirect trust relationships missing. This absence of information hinders decision makers from fully considering all possible trust data when making group decisions, thereby affecting the effectiveness of decision-making and the ability to reach consensus.

The decision-making environment in the real world is usually more complex, involving not only multiple trust transmission paths for decision makers but also mutual interference effects among these paths. To address this complexity, quantum probability theory is introduced in this paper. For each trust transfer chain, the indirect trust level is calculated using probability amplitude. Each trust transfer path is treated as a component of quantum states, with the mutual interference effect of each path reflected by the superposition and interference of quantum states. This approach more effectively addresses the problem of trust sequence transmission in trust networks. The process of completing indirect trust relationships is illustrated in Algorithm 1.

**Algorithm 1** A trust relationship aggregation method considering quantum interference.

- 1: **Input:** Initial trust relationship matrix  $T$ , where each element represents the direct trust relationship between decision-makers.
- 2: **Output:** Updated trust relationship matrix  $T'$ .
- 3: **Begin:**
- 4: **Step 1:** For each pair of decision-makers, check if there is no direct trust relationship or the trust value needs to be updated.
- 5: **Step 2:** Identify all trust transmission paths between the two decision-makers and record these paths.
- 6: **Step 3:** For each identified trust transmission path, calculate its trust contribution based on quantum probability theory.
- 7: **Step 4:** Aggregate all trust contributions from the identified paths by considering mutual interference effects according to quantum interference principles to obtain the final trust value.
- 8: **Step 5:** Update the corresponding element in the trust relationship matrix with the aggregated trust value.
- 9: **Step 6:** Repeat Steps 1–5 for all pairs of decision-makers in the trust network.
- 10: **Step 7:** Output the updated trust relationship matrix  $T'$ .
- 11: **End.**

Firstly, this paper adopts fuzzy trust matrix [36] to represent the trust relationship between decision makers, extending the traditional 0–1 trust relationship to any real number within the interval [0,1]. Trust matrix  $T = (t_{kh})_{s \times s}, t_{kh} \in [0, 1]$  is used to represent the degree of trust between decision makers, and  $e_{kh} (k = 1, 2, \dots, s, h = 1, 2, \dots, s, k \neq h)$  represents the path between decision makers  $d_k$  and  $d_h$  in the trust network, corresponding to the edges of the trust network.

Secondly, construct QLBN. Assume that there are  $q$  trust transmission paths in the trust network.  $E_r (r = 1, 2, \dots, q)$  indicates the direct trust path  $r - th (r = 1, 2, \dots, q)$  in the trust network. For example,  $E_1$  indicates the path between decision maker  $d_1$  and  $d_2$ . The first layer of the quantum-like Bayesian network is the trust path weight:  $p(E_r) = w_r, r = 1, 2, \dots, q$ , and the second layer is the relative indirect trust probability of each path.

$$p(T_r|E_r) = \frac{t_r}{\sum_{r=1}^q t_r}, r = 1, 2, \dots, q. \quad (12)$$

where,  $T_r (r = 1, 2, \dots, q)$  denotes the  $r$ -th indirect trust path and  $t_r (r = 1, 2, \dots, q)$  denotes the trust degree of the  $r$ th path.

Finally, the comprehensive indirect confidence is calculated based on quantum interference between paths.

$$t_{kh} = \sum_{r=1}^q p(E_r) p(T_r|E_r) + 2 \sum_{r=1}^{y-1} \sum_{r'=r+1}^y \sqrt{p(E_r) p(T_r|E_r) p(E_{r'}) p(T_{r'}|E_{r'})} \cos \zeta_{rr'}, \quad (13)$$

$$k = 1, 2, \dots, s, h = 1, 2, \dots, s.$$

### 3.2.2. The weight calculation of decision makers based on trust networks

According to the complementary complete trust matrix  $T = (t_{kh})_{s \times s}$ ,  $t_{kh} \in [0, 1]$ , the degree centrality of each decision-maker is calculated, and the comprehensive relative out-degree centrality  $COD_k$  ( $k = 1, 2, \dots, s$ ) of decision-maker  $d_k$  ( $k = 1, 2, \dots, s$ ) is defined as [38]:

$$COD_k = \frac{TOD_k \times TID_k}{\max(TOD_k \times TID_k)}, k = 1, 2, \dots, s. \quad (14)$$

The larger the comprehensive relative out-degree centrality  $COD_k$  ( $k = 1, 2, \dots, s$ ) is, the more likely decision-maker  $d_k$  is to trust other decision-makers. The trust in degree  $TID_k$  ( $k = 1, 2, \dots, s$ ) represents the sum of the trust degree of other nodes pointing to the trust path of this node, and the trust out degree  $TOD_k$  ( $k = 1, 2, \dots, s$ ) represents the sum of the trust degree of this node pointing to the trust path of other nodes.

$$TID_k = \sum_{k=1, k \neq h}^s t_{hk}, k = 1, 2, \dots, s. \quad (15)$$

$$TOD_k = \sum_{k=1, k \neq h}^s t_{kh}, h = 1, 2, \dots, s. \quad (16)$$

On this basis, the weight of decision makers is calculated as follows:

$$\lambda_k = \frac{COD_k}{\sum_{k=1}^s COD_k}, k = 1, 2, \dots, s. \quad (17)$$

### 3.2.3. Minimum adjustment group consensus optimization model considering trust relationship

In the process of MAGDM, achieving a high level of group consensus is an important factor to ensure decision quality and group satisfaction [37]. On the premise of not losing generality, this paper uses the weighted average (OWA) operator to calculate the collective decision matrix  $Z^o = (z_{ij}^o)_{m \times n}$ .

$$z_{ij}^o = \sum_{k=1}^s \lambda_k z_{ij}^k, i = 1, 2, \dots, m, j = 1, 2, \dots, n. \quad (18)$$

Apply Manhattan distance to calculate individual consensus level:

$$ICL(d_k) = 1 - \frac{\sum_{i=1}^m \sum_{j=1}^n |z_{ij}^k - z_{ij}^o|}{mn}, k = 1, 2, \dots, s. \quad (19)$$

Therefore, the group consensus level can be obtained by aggregating individual consensus level with OWA operator.

$$GCL = \sum_{k=1}^s \lambda_k ICL(d_k). \quad (20)$$

However, due to differences in preferences and cognitive limitations among decision makers, reaching consensus often requires some adjustments. Traditional group consensus models primarily focus on the direct adjustment of preferences, but in practical applications, decision makers are often constrained by their confidence in adjusting their ratings. Additionally, the trust relationships within the trust network significantly impact the adjustment process. To address these challenges, a group consensus optimization model based on trust relationships is proposed. This model introduces the confidence constraints

of decision makers on their evaluation values and integrates the influence of trust levels to achieve a pre-set group consensus threshold while minimizing adjustment costs. The model  $M_1$  as follows.

$$M_1 : \text{Min} \sum_{k=1}^s \sum_{i=1}^m \sum_{j=1}^n \left( |z_{ij}^k - z_{ij}^{*k}| + \vartheta \delta_{ij}^k \right) \quad (21)$$

$$\delta_{ij}^k = \begin{cases} 1, & \text{if } |z_{ij}^k - z_{ij}^{*k}| > b_{ij}^k \text{ and } COD_k < \overline{COD}_k \\ 0, & \text{if } |z_{ij}^k - z_{ij}^{*k}| \leq b_{ij}^k, \end{cases} \quad (21-1)$$

$$\overline{COD}_k = \frac{1}{s} \sum_{k=1}^s COD_k, \quad (21-2)$$

$$s.t. \begin{cases} 1 - \frac{1}{smn} \sum_{k=1}^s \sum_{i=1}^m \sum_{j=1}^n |z_{ij}^k - z_{ij}^o| = CL^*, & (21-3) \\ z_{ij}^o = \sum_{k=1}^s \lambda_k z_{ij}^k, & (21-4) \\ b_{ij}^k \in [0, 1], z_{ij}^{*k} \in [0, 1], \vartheta \geq 0, & (21-5) \\ i = 1, 2, \dots, m; j = 1, 2, \dots, n; k = 1, 2, \dots, s. & (21-6) \end{cases}$$

where  $\vartheta$  ( $\vartheta \geq 0$ ) represents the additional adjustment cost incurred when the adjustments exceed the acceptable threshold, and its value should be a non-negative real number.  $\delta_{ij}^k$  ( $i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$ ) represents the indicator function to determine whether additional adjustment costs need to be introduced, and  $b_{ij}^k$  ( $i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$ ) represents the confidence level of each decision maker's assessment of the opinion.  $CL^*$  ( $CL^* \in [0, 1]$ ) represents the optimal level of group consensus. The closer  $CL^*$  is to 1, the higher the level of group consensus.

In this model, the objective function represents the realization of optimal group consensus by minimizing adjustment costs. Constraints (21-1) and constraints (21-2) indicate that if the adjustment is outside the confidence interval, the trust level of the decision maker is checked, the adjustment is allowed only if the trust level is below average, and additional adjustment costs are introduced. Constraints (21-3) and (21-4) represent the optimal group consensus level, and constraints (21-5)–(21-6) are the interval range of each parameter.

**Theorem 1.** There is an optimal solution to model  $M_1$ .

**Proof.** Firstly, the objective function is a linear combination of variable  $\delta_{ij}^k$  ( $i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$ ). Since the absolute value part and  $\delta_{ij}^k$  ( $i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$ ) are both defined as convex functions, and the absolute value function is continuously differentiable in the domain of definition, the objective function is piecewise linear and convex. Secondly, the constraints are linear constraints or conditional constraints, and since these constraints are linear or piecewise linear, and the value range of variable  $z_{ij}^{*k}$  ( $i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$ ) is finite, the constraint set is a convex set. In summary, the objective function is convex and the constraints form a convex set. Therefore, according to the convex optimization theory [46], if the problem is a convex optimization problem and there is a feasible solution, then the problem has a global optimal solution.  $\square$

### 3.2.4. The weight calculation of attributes of decision-makers' risk preference

In MAGDM under uncertainty, a decision maker's choice is often influenced by incomplete information and potential risks. To more comprehensively simulate the risk preference of decision makers in the decision-making process, this paper constructs a multi-attribute risk utility function, which is defined as follows:

$$U_i = EXP \left( \sum_{j=1}^n \omega_j z_{ij}^o + \varepsilon_i \right), i = 1, 2, \dots, m. \quad (22)$$

where,  $\varepsilon_i$  ( $i = 1, 2, \dots, m$ ) is an error term conforming to normal distribution, representing the uncertainty in the real environment (such as changes in the external environment, loss of decision information, etc.).

**Theorem 2.** Multi-attribute risk utility functions  $U_i$  are convex functions.

**Proof.** Let  $f(x) = \exp(x)$ , which is a classical convex function. Its convexity can be proved by verifying its second derivative:  $f''(x) = \exp(x) > 0 \quad \forall x \in \mathbb{R}$ . Therefore,  $f(x)$  is a convex function. Let  $g(x) = \sum_{j=1}^n \varpi_j z_{ij}^o + \varepsilon_i$ , which is an affine function. An affine function is a special form of a linear function that keeps its convexity unchanged. By the combinatorial property of convex functions, if  $f(x)$  is convex and  $g(x)$  is affine, then the compound function  $f(g(x))$  is also convex. Therefore, the utility function  $U_i$  can be expressed as:  $U_i = f(g(x)) = \exp\left(\sum_{j=1}^n \varpi_j z_{ij}^o + \varepsilon_i\right)$ . According to the above properties,  $U_i$  is a convex function.  $\square$

**Conclusion 1.**  $U_i$  increases rapidly for larger utility values and can amplify the difference of utility values, thus distinguishing different alternative s more obviously.

**Conclusion 2.** Since  $U_i$  is a convex function, it indicates that this utility function exhibits risk preference characteristics. Specifically, with the increase of  $z_{ij}^o$ , the group's degree of risk preference increases. This means that the group tends to choose options with higher uncertainty in order to maximize their expected utility.

In uncertain multi-attribute group decision-making problems, the importance of each attribute is not only influenced by the decision-makers' subjective preference differences but also affected by external environmental changes and incomplete information. As a result, attributes' weights exhibit inherent uncertainty and variability. The Beta distribution, defined on the interval [0,1], is a flexible and adjustable continuous probability distribution that can simulate various forms of uncertainty by tuning its shape parameters, such as uniform, skewed, or bimodal distributions. Therefore, this paper employs the Beta distribution to model attribute weights, enabling a more realistic representation of their dynamic fluctuations and the intertwined subjective-objective uncertainty in complex decision-making environments. On this basis, this paper combines the risk preference function to simulate the risk preference that decision makers may face when weighing different options, and constructs the weight optimization model of attributes, as follows:

$$\begin{aligned} \text{Max } U_i &= EXP\left(\sum_{j=1}^n \varpi_j z_{ij}^o + \varepsilon_i\right), i = 1, 2, \dots, m \\ \text{s.t. } &\begin{cases} \sum_{j=1}^n \varpi_j = 1 \\ \varpi_j \sim \text{Beta}(a, b) \\ \varepsilon_i \sim N(0, \sigma^2) \\ \text{Beta}(\mu|a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \mu^{a-1} (1-\mu)^{b-1} \\ \Gamma(a) = \int_0^{+\infty} t^{a-1} e^{-t} dt \\ \varpi_j \in [0, 1], \alpha \geq 0, \beta \geq 0 \end{cases} \end{aligned} \quad (23)$$

In order to better capture the influence of random fluctuations and small perturbations on the decision results, the error term is made to follow the normal distribution. By solving model (23), the maximum risk appetite utility value of single alternative is obtained, denoted as  $U_i^*$  ( $i = 1, 2, \dots, m$ ), and the following single-objective optimization

model  $M_2$  is constructed to obtain the optimal solution.

$$\begin{aligned} M_2 : \text{Max } &\sum_{i=1}^m \frac{U_i}{U_i^*} \\ \text{s.t. } &\begin{cases} \sum_{j=1}^n \varpi_j = 1 & (24-1) \\ U_i = EXP\left(\sum_{j=1}^n \varpi_j z_{ij}^o + \varepsilon_i\right) & (24-2) \\ \varpi_j \sim \text{Beta}(a, b) & (24-3) \\ \varepsilon_i \sim N(0, \sigma^2) & (24-4) \\ \text{Beta}(\mu|a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \mu^{a-1} (1-\mu)^{b-1} & (24-5) \\ \Gamma(a) = \int_0^{+\infty} t^{a-1} e^{-t} dt & (24-6) \\ \varpi_j \in [0, 1], \alpha \geq 0, \beta \geq 0 & (24-7) \\ i = 1, 2, \dots, m; j = 1, 2, \dots, n & (24-8) \end{cases} \end{aligned} \quad (24)$$

The objective function of model is to maximize the comprehensive utility value of all alternatives, the constraint condition (24-1) ensures that the weight sum of attributes is 1, the constraint condition (24-2) represents the calculation of the comprehensive utility value of all alternatives, and the constraint condition (24-3)–(24-6) represents the distribution of each parameter. Constraints (24-7)–(24-8) limit the range of intervals for each parameter.

In addition, directly determining weights based on a single Beta distribution cannot fully capture environmental randomness and the stability of evaluation results. To address this, Monte Carlo simulation is introduced to generate a large number of random samples, simulating the possible variations and distribution characteristics of attribute weights under different scenarios. This approach significantly enhances the robustness and interpretability of the model outcomes. The detailed implementation process is provided in Algorithm 2.

**Algorithm 2** An attributes' weights optimization method based on Monte Carlo simulation.

- 1: **Input:** Decision matrix  $Z = (z_{ij}^o)_{m \times n}$ , Beta distribution parameters  $a$  and  $b$ , Normal distribution parameters  $\mu$  and  $\sigma$ , Number of simulations  $N$ .
- 2: **Output:** Optimal attributes' weights  $\varpi_j$  ( $j = 1, 2, \dots, n$ ).
- 3: **Begin:**
- 4: **Step 1:** Initialize parameters and variables.
- 5: **Step 2:** Perform Monte Carlo simulations for  $N$  iterations:
- 6: **for** each simulation **do**
- 7:   Generate attribute weights from the Beta distribution with parameters  $a$  and  $b$ .
- 8:   Normalize the generated attributes' weights so that their sum equals 1.
- 9:   Generate random error terms from the Normal distribution with parameters 0 and  $\sigma$ .
- 10:   Calculate the utility value of each alternative by combining attributes' weights, decision matrix values, and random errors, taking into account the decision-makers' risk preference function.
- 11: **end for**
- 12: **Step 3:** After completing all simulations, determine the maximum utility value for each alternative across all scenarios.
- 13: **Step 4:** Compute the objective function values for each simulation based on the aggregated maximum utility values.
- 14: **Step 5:** Identify the simulation that achieves the maximum objective function value and select the corresponding attribute weights as the final optimal weights.
- 15: **End.**

### 3.3. Ranking of alternatives based on score value

When the group consensus decision matrix and the optimal attribute weights are calculated, the final ranking of each alternative is obtained according to the score value.

$$Q_i = \sum_{j=1}^n \varpi_j z_{ij}^o, i = 1, 2, \dots, m. \quad (25)$$

### 3.4. Decision steps

In this paper, a multi-attribute quantum group consensus model considering interference effect based on trust network is proposed. The decision-making steps are as follows:

**Step 1:** Determine the evaluation information. Each decision maker  $d_k (k = 1, 2, \dots, s)$  evaluates the attributes  $c_j (j = 1, 2, \dots, n)$  in the alternative  $a_i (i = 1, 2, \dots, m)$  and obtains the information evaluation matrix  $Z^k = (z_{ij}^k)_{m \times n} (k = 1, 2, \dots, s)$ .

**Step 2:** Transform the initial information evaluation matrix into a decision matrix. According to Eqs. (8)–(10), the evaluation information matrix is transformed into a crisp number decision matrix  $\hat{Z}^k = (\hat{z}_{ij}^k)_{m \times n} (k = 1, 2, \dots, s)$ .

**Step 3:** Trust aggregation based on quantum interference. Considering the quantum interference complete indirect trust relationship between decision makers, the weights of decision makers is calculated according to Eqs. (12)–(17).

**Step 4:** Consider the minimum adjustment group consensus adjustment for trust relationships. Firstly, the group consensus level is initially calculated according to Eqs. (18)–(20). If the group consensus level is lower than the threshold value  $\kappa (\kappa \in [0, 1])$ , the feedback adjustment matrix is obtained by solving the model Eq. (21). Each decision maker makes the feedback adjustment according to the feedback matrix, and the group consensus level is calculated again, and the final decision matrix of each decision maker is obtained until the group consensus threshold is met.

**Step 5:** Calculate attributes' weights based on decision-maker's risk preference psychology. The optimal attribute weights were obtained by solving the model Eq. (24) through Monte Carlo simulation.

**Step 6:** Rank the alternatives. According to Eq. (25), the score values of alternatives are calculated and ranked.

## 4. Case analysis

### 4.1. Background

In the current global environment characterized by high uncertainty, intensifying international market competition, and the rapid advancement of Industry 4.0, intelligent manufacturing, and digital transformation, consumer electronics manufacturers are facing unprecedented challenges and opportunities. On one hand, emerging technologies such as artificial intelligence, big data, and the Internet of Things are accelerating the transformation of production models and industrial chains. On the other hand, global supply chain fluctuations, rising raw material costs, and increasingly diversified market demands are placing higher requirements on enterprises in terms of production efficiency, product quality, flexible manufacturing, and cost control. Against this backdrop, a leading international electronics manufacturing company has decided to comprehensively upgrade its core product production line to enhance overall competitiveness and market responsiveness. Although the current production line has performed well in recent years, with the accelerating pace of technological innovation and the growing demand for customized products, the existing line has become insufficient in terms of efficiency, flexibility, and resource allocation to meet the company's strategic development goals for the next three to five years. Therefore, the company plans to introduce advanced production technologies within the next six months and must select the optimal production line configuration from multiple solutions provided by various suppliers. However, the complexity and uncertainty of this selection process are exacerbated by differing priorities among decision-makers: technical experts focus on technological applicability and future scalability, financial managers emphasize initial investment and long-term operating costs, while senior management is concerned with strategic alignment and risk control.

In this context, the company formed a decision-making group composed of senior managers, technical experts, and financial advisors.

By adopting a multi-attribute group decision-making approach, the group systematically evaluated and compared each candidate solution across multiple dimensions. The objective was to achieve an optimal balance among technological feasibility, economic rationality, and strategic alignment, ultimately selecting the most suitable production line configuration that supports the company's long-term development strategy.

During the decision-making process, the team integrated four key attributes critical to the success of the line upgrade project: (1) Technical Suitability  $c_1$ : This attribute evaluates the adaptability of each solution to the existing production environment. High technical applicability means that solutions can be seamlessly integrated into existing systems, reducing implementation barriers and quickly yielding the desired results, thus ensuring a smooth upgrade process. (2) Implementation Feasibility  $c_2$ : This attribute reflects the feasibility of each solution given the enterprise's existing resources and capabilities. A solution with high implementation feasibility indicates that the enterprise has sufficient resources and capabilities to support the project's smooth development, thereby reducing implementation risks and improving the likelihood of success. (3) Future Scalability  $c_3$ : This attribute measures the potential of a production line configuration for future expansion and upgrades. High scalability means that the solution can be flexibly adjusted and upgraded in response to market demands and technological advancements, ensuring its long-term value and enhancing the company's competitiveness in the future market. (4) Cost-Effectiveness  $c_4$ : This attribute assesses the balance between cost inputs and benefit outputs for each option. Cost-effective solutions significantly improve production efficiency or reduce operating costs at a reasonable expense, thereby delivering higher economic returns to the enterprise and ensuring maximum return on investment for the project.

Based on these attributes, the enterprise evaluated four alternatives:

(1) Intelligent Fully Automated Production Line  $a_1$ : Provided by a leading German industrial automation company, this production line is fully automated, covering the entire process from raw material handling to finished product packaging. The solution represents the most cutting-edge technology in the current manufacturing industry and can significantly improve production efficiency and product quality. However, its initial investment cost is high, and maintenance costs are moderate.

(2) Modular Production Line  $a_2$ : Offered by a well-known manufacturing equipment company in Japan, this modular design allows for flexible combinations and rapid adjustments to the production line configuration, enabling it to meet the production needs of various products. Although the technology is not as advanced as the fully automated solution, the production flexibility is extremely high, and both the initial investment cost and maintenance costs are moderate.

(3) Intelligent Hybrid Production Line  $a_3$ : Developed by an emerging technology company in the United States, this production line combines the advantages of automation and manual operation, offering high technical advancement while retaining flexibility to adapt to the production of different products. The initial investment cost and maintenance costs are moderate, making it a balance between technological innovation and economic benefits.

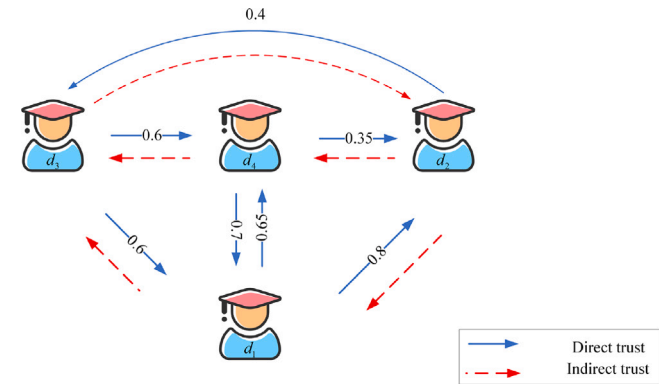
(4) Improved Traditional Production Line  $a_4$ : Provided by a long-established equipment manufacturer in China, this option focuses on enhancing the automation level of key processes in the existing production line while retaining the flexibility of manual operations. The initial investment cost is the lowest, but the technical advancement is limited, and maintenance costs are high.

### 4.2. Decision steps

Step 1: Collect decision maker evaluation information. The two linguistic sets of this example are  $A = \{low, middle, high\}$  and  $B = \{unlikely, fair, likely\}$ . On this basis, the initial evaluation information  $Z^k = (z_{ij}^k)_{4 \times 4} (k = 1, 2, \dots, 5)$  of each decision maker is collected. The

**Table 2**Initial evaluation information matrix  $Z^1$  for decision maker  $d_1$ .

|       | $c_1$                                                                                                                                                                         | $c_2$                                                                                                                                                                         |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $a_1$ | $(\sqrt{0.1e^{i\theta_{13}}} A_1\rangle + \sqrt{0.9e^{i\theta_{13}}} A_3\rangle) \otimes (\sqrt{0.35e^{i\theta_{23}}} B_2\rangle + \sqrt{0.65e^{i\theta_{23}}} B_3\rangle)$   | $(\sqrt{0.6e^{i\theta_{12}}} A_1\rangle + \sqrt{0.4e^{i\theta_{12}}} A_2\rangle) \otimes (\sqrt{0.35e^{i\theta_{12}}} B_1\rangle + \sqrt{0.65e^{i\theta_{12}}} B_2\rangle)$   |
| $a_2$ | $(\sqrt{0.4e^{i\theta_{13}}} A_1\rangle + \sqrt{0.6e^{i\theta_{13}}} A_3\rangle) \otimes (\sqrt{0.47e^{i\theta_{23}}} B_2\rangle + \sqrt{0.53e^{i\theta_{23}}} B_3\rangle)$   | $(\sqrt{0.7e^{i\theta_{12}}} A_2\rangle + \sqrt{0.3e^{i\theta_{12}}} A_3\rangle) \otimes (\sqrt{0.6e^{i\theta_{23}}} B_2\rangle + \sqrt{0.4e^{i\theta_{23}}} B_3\rangle)$     |
| $a_3$ | $(\sqrt{0.7e^{i\theta_{12}}} A_1\rangle + \sqrt{0.3e^{i\theta_{12}}} A_2\rangle) \otimes (\sqrt{0.6e^{i\theta_{12}}} B_1\rangle + \sqrt{0.4e^{i\theta_{12}}} B_2\rangle)$     | $(\sqrt{0.36e^{i\theta_{23}}} A_2\rangle + \sqrt{0.64e^{i\theta_{23}}} A_3\rangle) \otimes (\sqrt{0.4e^{i\theta_{12}}} B_1\rangle + \sqrt{0.6e^{i\theta_{12}}} B_3\rangle)$   |
| $a_4$ | $(\sqrt{0.5e^{i\theta_{12}}} A_1\rangle + \sqrt{0.5e^{i\theta_{12}}} A_2\rangle) \otimes (\sqrt{0.5e^{i\theta_{12}}} B_2\rangle + \sqrt{0.5e^{i\theta_{12}}} B_3\rangle)$     | $(\sqrt{0.45e^{i\theta_{12}}} A_1\rangle + \sqrt{0.55e^{i\theta_{12}}} A_2\rangle) \otimes (\sqrt{0.2e^{i\theta_{12}}} B_1\rangle + \sqrt{0.8e^{i\theta_{12}}} B_2\rangle)$   |
|       | $c_3$                                                                                                                                                                         | $c_4$                                                                                                                                                                         |
| $a_1$ | $(\sqrt{0.99e^{i\theta_{12}}} A_1\rangle + \sqrt{0.01e^{i\theta_{12}}} A_2\rangle) \otimes (\sqrt{0.65e^{i\theta_{12}}} B_2\rangle + \sqrt{0.35e^{i\theta_{12}}} B_3\rangle)$ | $(\sqrt{0.4e^{i\theta_{12}}} A_1\rangle + \sqrt{0.6e^{i\theta_{12}}} A_3\rangle) \otimes (\sqrt{0.85e^{i\theta_{23}}} B_1\rangle + \sqrt{0.15e^{i\theta_{23}}} B_3\rangle)$   |
| $a_2$ | $(\sqrt{0.7e^{i\theta_{12}}} A_1\rangle + \sqrt{0.3e^{i\theta_{12}}} A_2\rangle) \otimes (\sqrt{0.5e^{i\theta_{12}}} B_2\rangle + \sqrt{0.5e^{i\theta_{12}}} B_3\rangle)$     | $(\sqrt{0.7e^{i\theta_{12}}} A_2\rangle + \sqrt{0.3e^{i\theta_{12}}} A_3\rangle) \otimes (\sqrt{0.9e^{i\theta_{23}}} B_3\rangle + \sqrt{0.1e^{i\theta_{23}}} B_1\rangle)$     |
| $a_3$ | $(\sqrt{0.5e^{i\theta_{23}}} A_1\rangle + \sqrt{0.5e^{i\theta_{23}}} A_2\rangle) \otimes (\sqrt{0.5e^{i\theta_{13}}} B_1\rangle + \sqrt{0.5e^{i\theta_{13}}} B_3\rangle)$     | $(\sqrt{0.9e^{i\theta_{12}}} A_3\rangle + \sqrt{0.1e^{i\theta_{12}}} A_2\rangle) \otimes (\sqrt{0.7e^{i\theta_{12}}} B_3\rangle + \sqrt{0.3e^{i\theta_{12}}} B_1\rangle)$     |
| $a_4$ | $(\sqrt{0.46e^{i\theta_{23}}} A_2\rangle + \sqrt{0.54e^{i\theta_{23}}} A_3\rangle) \otimes (\sqrt{0.28e^{i\theta_{12}}} B_1\rangle + \sqrt{0.72e^{i\theta_{12}}} B_2\rangle)$ | $(\sqrt{0.34e^{i\theta_{13}}} A_1\rangle + \sqrt{0.66e^{i\theta_{13}}} A_3\rangle) \otimes (\sqrt{0.75e^{i\theta_{12}}} B_2\rangle + \sqrt{0.25e^{i\theta_{12}}} B_3\rangle)$ |

**Fig. 2.** Trust relationships among decision makers.

initial evaluation information of decision makers is shown in Table 2, and the rest is shown in Tables A.1–A.3 in Appendix.

Step 2: Transform the initial information evaluation matrix into a decision matrix. According to Eqs. (8)–(10), the evaluation information matrix of each decision maker is transformed into a crisp number decision matrix  $\hat{Z}^k = (\hat{z}_{ij}^k)_{4 \times 4}$  ( $k = 1, 2, \dots, 4$ ), as shown in Table 3.

Step 3: Consider the trust aggregation of quantum interference. The trust relationship between decision makers in this paper and the degree of trust on the network are shown in Fig. 2.

According to the trust network between decision makers, the direct trust matrix between decision makers can be obtained, where “ $\times$ ” indicates the indirect trust relationship between two decision makers. It is assumed that in this trust network, the trust relationship is passed sequentially, for example, in the trust relationship chain  $d_3 \rightarrow d_1 \rightarrow d_2$ . Trust transfer can occur sequentially only when the trust relationship between decision maker  $d_3$  to  $d_1$  and decision maker  $d_1$  to  $d_2$  exists at the same time. According to quantum probability theory, the degree of indirect trust between any two decision makers in the chain of trust transfer can be deduced. Since there may be multiple trust transfer paths between any two decision makers, the final indirect trust between

two decision makers can be calculated by considering the interference relationship between the trust paths. Assuming that the interference angle between paths is  $\xi_{rr'} = \pi/4$ , the trust path weight is reshown in Table A.4, and the trust network can be completed using Eqs. (12)–(13) as follows:

$$T = \begin{bmatrix} - & 0.8 & \times & 0.7 \\ \times & - & 0.4 & \times \\ 0.6 & \times & - & 0.6 \\ 0.65 & 0.35 & \times & - \end{bmatrix} \rightarrow T' = \begin{bmatrix} - & 0.8 & 0.7531 & 0.7 \\ 0.7609 & - & 0.4 & 0.3840 \\ 0.6 & 0.6761 & - & 0.6 \\ 0.65 & 0.35 & 0.7721 & - \end{bmatrix}.$$

According to the complete trust network matrix, the weights of each decision maker are calculated according to Eqs. (14)–(17).

$$A = \{0.3428, 0.2023, 0.2590, 0.2140\}.$$

Step 4: Consider the minimum adjustment group consensus adjustment for trust relationships. To maintain generality and facilitate model computation and result interpretation, the parameter  $\theta$  is set to 1 in the case analysis. The initial group consensus level  $GLC = 0.8556$  is calculated according to Eqs. (18)–(20). Due to the group consensus threshold  $\kappa = 0.9$ , it is necessary to solve the model  $M_1$  to obtain the feedback information matrix of each decision maker for reference and adjustment, as shown in Table 4. The bounded confidence matrices for each decision maker are shown in Table A.5 in the Appendix.

On this basis, each decision maker makes adjustments and obtains the decision matrix  $\hat{Z}^{k(1)} = (\hat{z}_{ij}^{k(1)})_{4 \times 4}$  of the first round of adjustments, as shown in Table 5.

At this point, the group consensus level is  $GLC = 0.8811 < 0.9$ , making it necessary to reorganize the decision-makers to adjust their opinions. The second round of feedback information matrix is presented in Table 6.

On this basis, each decision maker makes adjustments and obtains the decision matrix  $\hat{Z}^{k(2)} = (\hat{z}_{ij}^{k(2)})_{4 \times 4}$  of the second round of adjustments, as shown in Table 7.

The group consensus level after this round of adjustment is  $GLC = 0.9011 > 0.9$ , so the adjustment is stopped and the final group decision matrix  $Z^0 = (z_{ij}^0)_{4 \times 4}$  is obtained.

$$Z^0 = \begin{bmatrix} 0.8929 & 0.8980 & 0.8738 & 0.6577 \\ 0.6464 & 0.9268 & 0.8095 & 0.6145 \\ 0.8727 & 0.8694 & 0.8199 & 0.8681 \\ 0.6738 & 0.8988 & 0.8428 & 0.6940 \end{bmatrix}.$$

Step 5: Calculate the weights of attributes based on decision-maker's risk preference psychology. By definition of the Beta distribution, parameters  $a$  and  $b$  are positive. The calculation of  $a = 2$  and  $b = 5$  in this section reflects a strong bias, which is consistent with the actual situation in which enterprises have significant bias for some key attributes. The optimal attributes' weights were obtained by solving the model  $M_2$  through Monte Carlo simulation.

$$W = \{0.1028, 0.4683, 0.1962, 0.2327\}.$$

The model generates a large number of attribute weight combinations through Monte Carlo simulation and evaluates the utility value of each alternative under these weights. The box plot shown in Fig. 3 reflects the distribution of utility values for each alternative across different weight combinations. By examining these distributions, one can identify which combinations of weights yield a higher and more stable utility value for an alternative in most cases. The model selects the optimal weight combination by maximizing the combined utility values of all alternatives, ensuring that the distribution of these utility values performs best overall.

**Table 3**  
Crisp number decision matrix  $\hat{Z}^k (k = 1, 2, 3, 4)$  of decision makers  $d_k (k = 1, 2, 3, 4)$ .

|       | $c_1$  | $c_2$  | $c_3$  | $c_4$  |  | $c_1$  | $c_2$  | $c_3$  | $c_4$  |
|-------|--------|--------|--------|--------|--|--------|--------|--------|--------|
|       | $d_1$  |        |        |        |  | $d_2$  |        |        |        |
| $a_1$ | 1.0000 | 0.8253 | 1.0000 | 0.3606 |  | 0.5968 | 1.0000 | 0.5540 | 0.6240 |
| $a_2$ | 0.5838 | 0.8783 | 0.5646 | 0.4427 |  | 0.5492 | 0.8770 | 0.8293 | 0.5756 |
| $a_3$ | 0.6749 | 0.7973 | 0.6973 | 1.0000 |  | 0.8398 | 0.8832 | 1.0000 | 0.3865 |
| $a_4$ | 0.5594 | 1.0000 | 0.8731 | 0.3207 |  | 1.0000 | 0.6795 | 0.3718 | 1.0000 |
|       | $d_3$  |        |        |        |  | $d_4$  |        |        |        |
| $a_1$ | 0.9498 | 0.9666 | 1.0000 | 0.6580 |  | 0.8248 | 0.8289 | 0.3852 | 0.8146 |
| $a_2$ | 0.7459 | 1.0000 | 0.9634 | 0.4794 |  | 0.4679 | 1.0000 | 0.5372 | 1.0000 |
| $a_3$ | 1.0000 | 0.8869 | 0.9080 | 1.0000 |  | 1.0000 | 0.9078 | 0.6770 | 0.5257 |
| $a_4$ | 0.7230 | 0.9410 | 0.8531 | 0.6178 |  | 0.4579 | 0.8617 | 1.0000 | 0.8085 |

**Table 4**  
First round feedback matrix  $Z'^{k(1)} (k = 1, 2, 3, 4)$  of decision makers  $d_k (k = 1, 2, 3, 4)$ .

|       | $c_1$  | $c_2$  | $c_3$  | $c_4$  |  | $c_1$  | $c_2$  | $c_3$  | $c_4$  |
|-------|--------|--------|--------|--------|--|--------|--------|--------|--------|
|       | $d_1$  |        |        |        |  | $d_2$  |        |        |        |
| $a_1$ | 1.0000 | 0.9253 | 1.0000 | 0.6606 |  | 0.7468 | 1.0000 | 0.7540 | 0.9240 |
| $a_2$ | 0.7338 | 0.9783 | 0.7646 | 0.7427 |  | 0.6992 | 0.9770 | 1.0000 | 0.8756 |
| $a_3$ | 0.8249 | 0.8973 | 0.8973 | 1.0000 |  | 0.9898 | 0.9832 | 1.0000 | 0.6865 |
| $a_4$ | 0.7094 | 1.0000 | 1.0000 | 0.6207 |  | 1.0000 | 0.7795 | 0.5718 | 1.0000 |
|       | $d_3$  |        |        |        |  | $d_4$  |        |        |        |
| $a_1$ | 1.0000 | 1.0000 | 1.0000 | 0.9580 |  | 0.9748 | 0.9289 | 0.5852 | 1.0000 |
| $a_2$ | 0.8959 | 1.0000 | 1.0000 | 0.7794 |  | 0.6179 | 1.0000 | 0.7372 | 1.0000 |
| $a_3$ | 1.0000 | 0.9869 | 1.0000 | 1.0000 |  | 1.0000 | 1.0000 | 0.8770 | 0.8257 |
| $a_4$ | 0.8730 | 1.0000 | 1.0000 | 0.9178 |  | 0.6079 | 0.9617 | 1.0000 | 1.0000 |

**Table 5**  
First round adjustment matrix  $\hat{Z}^{k(1)} (k = 1, 2, 3, 4)$  of decision makers  $d_k (k = 1, 2, 3, 4)$ .

|       | $c_1$  | $c_2$  | $c_3$  | $c_4$  |  | $c_1$  | $c_2$  | $c_3$  | $c_4$  |
|-------|--------|--------|--------|--------|--|--------|--------|--------|--------|
|       | $d_1$  |        |        |        |  | $d_2$  |        |        |        |
| $a_1$ | 0.9453 | 0.8253 | 0.9805 | 0.4604 |  | 0.7952 | 1.0000 | 0.7450 | 0.7016 |
| $a_2$ | 0.5838 | 0.8783 | 0.7601 | 0.4427 |  | 0.5492 | 0.8770 | 0.8293 | 0.5756 |
| $a_3$ | 0.6749 | 0.8215 | 0.6973 | 1.0000 |  | 0.8398 | 0.8832 | 1.0000 | 0.3865 |
| $a_4$ | 0.5594 | 1.0000 | 0.8731 | 0.3207 |  | 0.9650 | 0.6795 | 0.5017 | 0.9452 |
|       | $d_3$  |        |        |        |  | $d_4$  |        |        |        |
| $a_1$ | 0.9598 | 0.9666 | 1.0000 | 0.6580 |  | 0.8248 | 0.8289 | 0.4816 | 0.8146 |
| $a_2$ | 0.7459 | 0.9659 | 0.9453 | 0.4794 |  | 0.5014 | 1.0000 | 0.7015 | 1.0000 |
| $a_3$ | 1.0000 | 0.8869 | 0.8873 | 1.0000 |  | 1.0000 | 0.9078 | 0.7543 | 0.6754 |
| $a_4$ | 0.7230 | 0.9410 | 0.8531 | 0.6178 |  | 0.4579 | 0.8617 | 1.0000 | 0.8085 |

**Table 6**  
Second round feedback matrix  $Z'^{k(2)} (k = 1, 2, 3, 4)$  of decision makers  $d_k (k = 1, 2, 3, 4)$ .

|       | $c_1$  | $c_2$  | $c_3$  | $c_4$  |  | $c_1$  | $c_2$  | $c_3$  | $c_4$  |
|-------|--------|--------|--------|--------|--|--------|--------|--------|--------|
|       | $d_1$  |        |        |        |  | $d_2$  |        |        |        |
| $a_1$ | 1.0000 | 0.9253 | 1.0000 | 0.7604 |  | 0.9452 | 1.0000 | 0.9450 | 1.0000 |
| $a_2$ | 0.7338 | 0.9783 | 0.9601 | 0.7427 |  | 0.6992 | 0.9770 | 1.0000 | 0.8756 |
| $a_3$ | 0.8249 | 0.9215 | 0.8973 | 1.0000 |  | 0.9898 | 0.9832 | 1.0000 | 0.6865 |
| $a_4$ | 0.7094 | 1.0000 | 1.0000 | 0.6207 |  | 1.0000 | 0.7795 | 0.7017 | 1.0000 |
|       | $d_3$  |        |        |        |  | $d_4$  |        |        |        |
| $a_1$ | 1.0000 | 1.0000 | 1.0000 | 0.9580 |  | 0.9748 | 0.9289 | 0.6816 | 1.0000 |
| $a_2$ | 0.8959 | 1.0000 | 1.0000 | 0.7794 |  | 0.6514 | 1.0000 | 0.9015 | 1.0000 |
| $a_3$ | 1.0000 | 0.9869 | 1.0000 | 1.0000 |  | 1.0000 | 1.0000 | 0.9543 | 0.9754 |
| $a_4$ | 0.8730 | 1.0000 | 1.0000 | 0.9178 |  | 0.6079 | 0.9617 | 1.0000 | 1.0000 |

Specifically, when a alternative's utility value has a higher median, less variability, and fewer outliers across different weight combinations, it indicates that the alternative can perform well under most weight combinations. This suggests that the corresponding weight combination can more reliably reflect the contribution of each attribute to the decision outcome. Consequently, the model can identify a set of optimal attribute weights through this analysis process, resulting in a more robust and effective final decision. In this way, the model effectively balances the importance of different attributes, thereby calculating an optimal attributes' weights combination.

Step 6: Rank the alternatives. According to Eq. (25), the score values of alternatives are calculated and ranked.

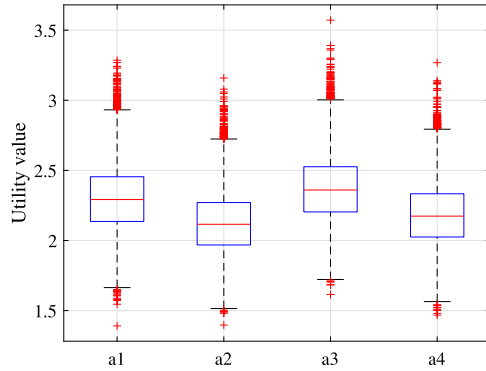
$$Q_1 = 0.8368, Q_2 = 0.8023, Q_3 = 0.8597, Q_4 = 0.8170.$$

Therefore, the final alternative ranking result is  $a_3 > a_1 > a_4 > a_2$ , that is, the intelligent hybrid production line  $a_3$  is selected as the production line configuration alternative suitable for the long-term strategic goals of the enterprise.

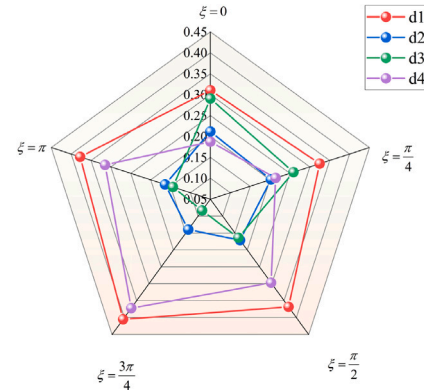
The intelligent hybrid production line  $a_3$  combines high technical advancement and good flexibility, and achieves a good balance between initial investment and long-term operating costs. It can adapt to

**Table 7**  
Second round adjustment matrix  $\hat{Z}^{k(2)} (k = 1, 2, 3, 4)$  of decision makers.

|       | $c_1$  | $c_2$  | $c_3$  | $c_4$  |  | $c_1$  | $c_2$  | $c_3$  | $c_4$  |
|-------|--------|--------|--------|--------|--|--------|--------|--------|--------|
|       | $d_1$  |        |        |        |  | $d_2$  |        |        |        |
| $a_1$ | 0.9453 | 0.8253 | 0.9805 | 0.5267 |  | 0.7952 | 1.0000 | 0.8063 | 0.7016 |
| $a_2$ | 0.5838 | 0.8783 | 0.7601 | 0.4427 |  | 0.5861 | 0.8770 | 0.8293 | 0.6159 |
| $a_3$ | 0.7359 | 0.8215 | 0.6973 | 1.0000 |  | 0.8398 | 0.8832 | 1.0000 | 0.5000 |
| $a_4$ | 0.5594 | 1.0000 | 0.8731 | 0.5230 |  | 0.9650 | 0.7216 | 0.6147 | 0.9452 |
|       | $d_3$  |        |        |        |  | $d_4$  |        |        |        |
| $a_1$ | 0.9598 | 0.9666 | 0.9765 | 0.6580 |  | 0.8248 | 0.8289 | 0.6514 | 0.8146 |
| $a_2$ | 0.7459 | 0.9659 | 0.9453 | 0.5102 |  | 0.6782 | 1.0000 | 0.7015 | 1.0000 |
| $a_3$ | 0.9650 | 0.8869 | 0.8873 | 1.0000 |  | 1.0000 | 0.9078 | 0.7543 | 0.8563 |
| $a_4$ | 0.7230 | 0.9410 | 0.8531 | 0.6178 |  | 0.5124 | 0.8617 | 1.0000 | 0.8085 |



**Fig. 3.** Box plot of utility values of each alternative based on Monte Carlo simulation.



**Fig. 4.** Influence of interference angle change on decision makers' weights.

rapid changes in the market, and is an ideal choice for both realistic feasibility and future development strategy.  $a_1$ , the smart fully automated production line in second place, is the most advanced technology, but the initial investment is high and the flexibility is slightly less, while suitable for the pursuit of extremely high automation scenarios, but for the current strategic goals of enterprises need to trade off between cost and flexibility. The third ranked modular production line  $a_2$  has strong flexibility, but the technology is less advanced, and it is difficult to occupy an advantage in high-end manufacturing scenarios. The improved traditional production line  $a_4$  ranks last, although it has the lowest cost, but the technical level is backward, and it is difficult to meet the long-term transformation and upgrading needs of enterprises. Therefore, the intelligent hybrid production line  $a_3$  as a long-term strategic goal of the production line configuration scheme is reasonable.

#### 4.3. Sensitivity analysis

##### (1) Sensitivity analysis of interference angle variation

In the case analysis, the interference angle is  $45^\circ$ , which represents the positive interference of medium intensity, so as to reflect that there is a certain degree of synergy between decision makers in the actual decision-making process, while retaining a certain degree of independence. In order to further verify the influence of interference angle setting on the model results, this paper systematically investigated the influence of interference angle on the weight distribution of decision makers at five typical angles of  $0^\circ$ ,  $45^\circ$ ,  $90^\circ$ ,  $135^\circ$  and  $180^\circ$  in the sensitivity analysis, as shown in Fig. 4.

When the interference angle between trust paths changes, it directly affects the final trust relationships among decision-makers, thereby influencing the distribution of decision-makers' weights. As shown in Fig. 4, when the direction of interference between trust paths shifts from positive to negative, the weights of decision-makers exhibit significant dynamic changes. Among them, decision-maker  $d_1$  maintains a

consistently high weight, indicating a strong core position and stability within the group decision-making process. In contrast, the weights of other decision-makers fluctuate considerably with changes in the interference angle, reflecting their higher susceptibility to interference effects and sensitivity to variations in the phase of trust paths, resulting in oscillations in their influence levels.

This is because when the interference angle is in the positive interference interval, the phase relationship between the trust paths strengthens the superposition effect of trust, and the weight of some decision makers remains relatively stable, indicating that these members have strong stability and dominance in the decision network. However, when the interference angle gradually enters the negative interference interval, there is a counteracting effect between the trust paths, which leads to a significant decrease in the weight of marginal or non-core decision-makers, and the fluctuation of influence of some decision-makers is intensified, reflecting the dynamic adjustment of the status of decision-makers with the change of phase. The results show that the change of the phase angle of the trust path has an important regulating effect on the weight distribution of the members in the group. Through the sensitivity analysis of interference angle, the key decision-makers with the greatest influence under different trust structure conditions can be identified, and the information can be used to adjust the decision-making process, so as to more effectively guide the formation of group consensus, and enhance the robustness of the model to the dynamic trust structure changes under complex uncertain environments.

##### (2) Parameter sensitivity analysis in Beta distribution

In a multi-attribute decision model, the beta distribution can be used to describe the distribution of attribute weights whose shape is controlled by the parameters  $a$  and  $b$ . Different combinations of  $a$  and  $b$  will lead to different weight distribution shapes, which will affect the comprehensive utility value of each alternative and thus affect the decision result, this paper sets  $a$  and  $b$  to vary within the interval

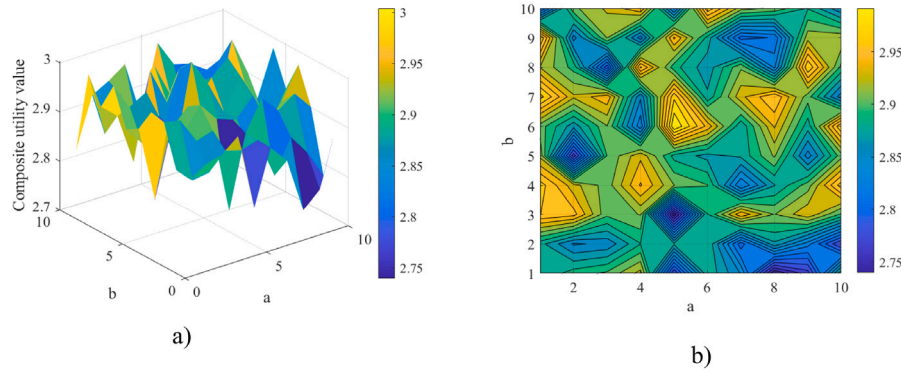


Fig. 5. Influence of different parameter combinations ( $a, b$ ) on the objective function.

Table 8

Attributes' weights under different ( $a, b$ ) combinations.

| $a$ | $b$ | $\varpi_1$ | $\varpi_2$ | $\varpi_3$ | $\varpi_4$ | $a$ | $b$ | $\varpi_1$ | $\varpi_2$ | $\varpi_3$ | $\varpi_4$ |
|-----|-----|------------|------------|------------|------------|-----|-----|------------|------------|------------|------------|
| 1   | 1   | 0.2400     | 0.2936     | 0.4600     | 0.0064     | 6   | 6   | 0.1873     | 0.3149     | 0.3217     | 0.1761     |
| 2   | 2   | 0.1454     | 0.3140     | 0.2509     | 0.2897     | 7   | 7   | 0.2812     | 0.2208     | 0.2460     | 0.2520     |
| 3   | 3   | 0.2568     | 0.2782     | 0.4098     | 0.0552     | 8   | 8   | 0.3240     | 0.3299     | 0.1790     | 0.1671     |
| 4   | 4   | 0.1499     | 0.3616     | 0.3639     | 0.1246     | 9   | 9   | 0.1874     | 0.2613     | 0.2446     | 0.3067     |
| 5   | 5   | 0.2954     | 0.3028     | 0.2794     | 0.1224     | 10  | 10  | 0.2831     | 0.2802     | 0.2252     | 0.2115     |

[1,10] respectively, systematically analyzes the influence of different parameter combinations on attribute weight distribution, as shown in Fig. 5.

On this basis, 10 groups of ( $a, b$ ) combinations are selected to observe the changes of attribute weights, as shown in Table 8.

As can be seen from Table 8, with the change of parameter group ( $a, b$ ), the weight of attributes will change significantly. For example, when  $a = b = 1$ , the weight of attribute  $c_3$  is significantly higher than that of the other attributes, while when  $a = b = 7$ , the weight of each attribute has a small difference and is more uniform.

Therefore, in practical applications, this strategy of influencing attribute weights by adjusting beta distribution parameters can help decision makers make more reasonable and flexible decisions in various decision scenarios. Depending on different preferences and situational requirements, decision makers can select appropriate parameter combinations to achieve optimal weight allocation for each attribute, thereby optimizing the decision outcome.

### (3) Other parameters

Although other parameters involved in this paper are not the core research object of the model, their Settings also follow the theoretical basis and practical application requirements, and have a clear basis and rationality.

Firstly, in the group consensus model, the confidence coefficient  $b_{ij}^k$  ( $i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$ ) of each decision maker is used to describe the reliability and stability of decision makers' opinions. This coefficient is usually derived from historical evaluation data, decision maker experience, and professional evaluation, and is part of the input criteria rather than a model tunable parameter, so sensitivity analysis is not required. Secondly, the additional adjustment cost factor  $\theta$  in the consensus optimization model is designed to balance the trade-off between the magnitude of adjustment and the cost of adjustment. For the sake of generality,  $\theta$  is usually set to 1 to reflect moderately adjusted penalty strength, avoid introducing artificial preferences, and facilitate comparison and interpretation of model results. In addition, in the multi-attribute risk-utility function proposed in this paper, the error term  $\varepsilon_i$  ( $i = 1, 2, \dots, m$ ) is used to simulate uncontrollable factors (such as changes in external environment, information loss or cognitive bias) in the real decision-making environment. Due to the normalization of modeling, this paper assumes that the error term follows a normal distribution. This hypothesis has strong theoretical support and is a

Table 9

Comparison of numerical experimental ranking results of different models.

| Method               | Ranking indices                                                                         | Final ranking orders    |
|----------------------|-----------------------------------------------------------------------------------------|-------------------------|
| TOPSIS [47]          | $C_1^* = 0.8322, C_2^* = 0.1695,$<br>$C_3^* = 0.8306, C_4^* = 0.3295$                   | $a_1 > a_3 > a_4 > a_2$ |
| TODIM [48]           | $\phi(x_1) = 0.3144, \phi(x_2) = -0.3579,$<br>$\phi(x_3) = 0.4043, \phi(x_4) = -0.3609$ | $a_3 > a_1 > a_2 > a_4$ |
| VIKOR [49]           | $Q(a_1) = 0.0000, Q(a_2) = 1.0000,$<br>$Q(a_3) = 0.6764, Q(a_4) = 0.5375$               | $a_1 > a_4 > a_3 > a_2$ |
| PROMETHEE [50]       | $\phi(x_1) = 0.0052, \phi(x_2) = -0.0178,$<br>$\phi(x_3) = 0.0205, \phi(x_4) = -0.0080$ | $a_3 > a_1 > a_2 > a_4$ |
| Quantum model 1 [32] | $p(a_1) = 0.3652, p(a_2) = 0.1631,$<br>$p(a_3) = 0.2775, p(a_4) = 0.1942$               | $a_1 > a_3 > a_4 > a_2$ |
| Quantum model 2 [30] | $p(a_1) = 0.2400, p(a_2) = 0.0025,$<br>$p(a_3) = 0.1597, p(a_4) = 0.5978$               | $a_4 > a_1 > a_3 > a_2$ |
| Quantum model 3 [31] | $p(a_1) = 0.1688, p(a_2) = 0.2638,$<br>$p(a_3) = 0.3476, p(a_4) = 0.2198$               | $a_3 > a_2 > a_4 > a_1$ |
| The proposed model   | $Q_1 = 0.8368, Q_2 = 0.8023,$<br>$Q_3 = 0.8597, Q_4 = 0.8170$                           | $a_3 > a_1 > a_4 > a_2$ |

common processing method in econometrics and behavioral decision models, which can better capture the influence of random volatility and small perturbations on decision results. Therefore, such parameters have been fully demonstrated in theory, and will not have a substantial impact on the stability of the model, and further sensitivity analysis is not necessary.

### 4.4. Comparative analysis

In order to evaluate the effectiveness and robustness of the proposed model, numerical experiments will be conducted to compare and analyze the model with other existing decision models. The final ranking results obtained by different methods are shown in Table 9.

As shown in Table 9, the ranking results of alternative schemes obtained by the proposed model show significant differences compared with traditional decision-making models. The core idea of the TOPSIS method [47] is to calculate the distance between each alternative and the ideal and negative-ideal solutions, selecting the alternative that is closest to the ideal solution and farthest from the negative-ideal solution. However, this method generally relies on subjectively assigned weights and ignores group interaction and interference effects among decision-makers. As a result, the TOPSIS method [47] identifies  $a_1$  as the optimal alternative, which differs from the result of the proposed model. The TODIM method [48], based on prospect theory, simulates decision-makers' psychological responses to gains and losses and emphasizes psychological perception and behavioral preferences in the ranking process. Although its optimal alternative is consistent with this paper, the ranking of other alternatives differs significantly. This discrepancy arises because the TODIM method [48] does not fully consider trust relationships and interference effects among decision-makers, making its results more susceptible to subjective influences.

In addition, the core idea of the VIKOR method [49] is to rank alternatives based on their closeness to the ideal solution and by minimizing the maximum individual regret, comprehensively considering both group satisfaction and individual satisfaction. However, this method also relies on subjectively assigned weights and does not model the trust structure and interference effects among decision-makers, resulting in significant differences in the ranking outcomes compared to the model proposed in this paper, with the optimal choice not being  $a_3$ . The PROMETHEE method [50], based on preference functions and the theory of positive and negative outranking flows, can address multi-criteria decision-making problems and to some extent reflect decision-makers' preference information. Although its optimal solution is consistent with this paper's result, the rankings of other alternatives differ noticeably. The main reason is that it does not incorporate dynamic trust relationship adjustments and interference effect modeling, leading to insufficient adaptability when dealing with weight fluctuations and the complexity of information interactions.

In comparison with existing quantum decision-making models, Quantum model 1 [32] employs quantum interference effects to capture nonlinear interactions among decision-makers. However, it lacks an objective and systematic mechanism for calculating decision-makers' and attributes' weights, relying solely on quantum probability-based information processing. This leads to the selection of  $a_1$  as the optimal alternative, differing from the results of this paper. Quantum model 2 [30] introduces a group consensus adjustment mechanism, which partially considers the consistency of group opinions. However, it does not incorporate trust relationships among decision-makers or model information uncertainty in complex environments, resulting in limited applicability to uncertain social network settings. The optimal alternative of Quantum model 3 [31] is the same as  $a_3$  in this paper, but there are obvious differences in the order of other schemes. The reason is that the model focuses on the modeling of interference effect, but does not fully construct the dynamic adjustment mechanism of group opinions, so that the ranking results fail to reflect the dynamic evolution process of opinion adjustment and consensus formation in the decision-making process.

In contrast, the model proposed in this paper aggregates trust relationships among decision-makers based on quantum probability theory and objectively calculates decision-makers' weights using the comprehensive relative out-degree centrality index. Furthermore, a minimum adjustment group consensus optimization model based on trust relationships is established to dynamically adjust group opinions and achieve group consensus. In addition, considering the risk preference psychology of decision-makers, Beta distribution and Monte Carlo simulation are introduced to construct an optimization model for dynamically determining the optimal attribute weights. Therefore, the results of the proposed model are more consistent with the complex realities of uncertain decision-making environments and demonstrate strong adaptability and robustness.

## 5. Conclusion

In this paper, the concept of quantum probability theory is introduced into the traditional LZN framework, extending the LZN to a quantum state form. This extension allows information representations to not only account for fuzziness and confidence but also to more accurately describe and quantify interference effects in the decision process through the superposition and interference of quantum states. Specifically, QLZN uses probabilistic amplitudes of quantum states to characterize complex interactions between information, enabling dynamic changes and uncertainties to be captured more accurately in linguistic information evaluation. On this basis, a novel QLZN multi-attribute group consensus model, based on quantum interference, is proposed. The model aggregates trust paths using quantum probability theory to complete the trust network and constructs a group consensus optimization model based on minimal adjustment, considering the

influence of trust degree to provide feedback and suggestions to help decision makers achieve group consensus. Additionally, the model takes into account decision-makers' risk preference and optimizes attributes' weights allocation by maximizing the comprehensive utility value of alternative solutions, combining Monte Carlo simulation. The results show that the interference angle between trust paths has a significant influence on the weight distribution of decision makers, and the change of Beta distribution parameters will change the distribution of attributes' weights, and then affect the final decision result. This finding not only validates the effectiveness of the proposed model in dealing with uncertainty and complex interactions, but also shows that quantum probability theory can provide theoretical support for describing the interference effects and dynamic evolution of information among complex groups in the decision-making process, thus enhancing the applicability and interpretation ability of the model in uncertain complex decision-making scenarios.

Although the model proposed in this paper is innovative and effective in solving uncertain multi-attribute group decision-making problems in social networks, there are still two limitations. First of all, the model in this paper is mainly aimed at static decision-making scenarios, and has not fully considered the dynamic influence of trust relationship and evaluation information evolution over time. Secondly, although the model is based on quantum probability theory, there are some limitations in theoretical application, especially in the precise measurement of interference angle. In a complex environment, the interference angle is difficult to be accurately quantified, which may affect the aggregation result of trust relationship, and then affect the application effect of the model. Therefore, future research will focus on two directions: On the one hand, extend the model to the dynamic social network environment, and deeply depict the evolution process of decision makers' behavior and trust relationship; On the other hand, the quantization method and precision calibration mechanism of interference angle are further explored to improve the applicability and robustness of the decision model based on quantum probability theory in practical complex situations.

## CRedit authorship contribution statement

**Yizhao Xu:** Writing – original draft, Methodology, Data curation. **Shuli Yan:** Writing – review & editing, Validation, Methodology, Conceptualization. **Francisco Javier Cabrerizo:** Writing – review & editing, Supervision, Conceptualization.

## Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, and we do not have any possible conflicts of interest.

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## Appendix

See [Tables A.1–A.5](#).

**Table A.1**Initial evaluation information matrix for decision makers  $d_2$ .

|       | $c_1$                                                                                                                                                                         | $c_2$                                                                                                                                                                         |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $a_1$ | $(\sqrt{0.21e^{i\theta_{12}} A_1\rangle} + \sqrt{0.79e^{i\theta_{12}} A_2\rangle}) \otimes (\sqrt{0.35e^{i\theta_{12}} B_1\rangle} + \sqrt{0.65e^{i\theta_{12}} B_2\rangle})$ | $(\sqrt{0.38e^{i\theta_{12}} A_1\rangle} + \sqrt{0.62e^{i\theta_{12}} A_3\rangle}) \otimes (\sqrt{0.12e^{i\theta_{23}} B_2\rangle} + \sqrt{0.88e^{i\theta_{23}} B_3\rangle})$ |
| $a_2$ | $(\sqrt{0.43e^{i\theta_{13}} A_1\rangle} + \sqrt{0.67e^{i\theta_{13}} A_3\rangle}) \otimes (\sqrt{0.38e^{i\theta_{23}} B_2\rangle} + \sqrt{0.62e^{i\theta_{23}} B_3\rangle})$ | $(\sqrt{0.49e^{i\theta_{12}} A_1\rangle} + \sqrt{0.51e^{i\theta_{12}} A_2\rangle}) \otimes (\sqrt{0.16e^{i\theta_{23}} B_2\rangle} + \sqrt{0.84e^{i\theta_{23}} B_3\rangle})$ |
| $a_3$ | $(\sqrt{0.1e^{i\theta_{23}} A_1\rangle} + \sqrt{0.9e^{i\theta_{23}} A_3\rangle}) \otimes (\sqrt{0.25e^{i\theta_{23}} B_2\rangle} + \sqrt{0.75e^{i\theta_{23}} B_3\rangle})$   | $(\sqrt{0.3e^{i\theta_{12}} A_1\rangle} + \sqrt{0.7e^{i\theta_{12}} A_2\rangle}) \otimes (\sqrt{0.76e^{i\theta_{12}} B_1\rangle} + \sqrt{0.24e^{i\theta_{12}} B_3\rangle})$   |
| $a_4$ | $(\sqrt{0.02e^{i\theta_{23}} A_1\rangle} + \sqrt{0.98e^{i\theta_{23}} A_3\rangle}) \otimes (\sqrt{0.24e^{i\theta_{13}} B_2\rangle} + \sqrt{0.76e^{i\theta_{13}} B_3\rangle})$ | $(\sqrt{0.66e^{i\theta_{12}} A_1\rangle} + \sqrt{0.34e^{i\theta_{12}} A_2\rangle}) \otimes (\sqrt{0.35e^{i\theta_{12}} B_1\rangle} + \sqrt{0.65e^{i\theta_{12}} B_2\rangle})$ |
|       | $c_3$                                                                                                                                                                         | $c_4$                                                                                                                                                                         |
| $a_1$ | $(\sqrt{0.5e^{i\theta_{23}} A_2\rangle} + \sqrt{0.5e^{i\theta_{23}} A_3\rangle}) \otimes (\sqrt{0.15e^{i\theta_{13}} B_3\rangle} + \sqrt{0.85e^{i\theta_{13}} B_2\rangle})$   | $(\sqrt{0.27e^{i\theta_{13}} A_1\rangle} + \sqrt{0.73e^{i\theta_{13}} A_3\rangle}) \otimes (\sqrt{0.19e^{i\theta_{23}} B_1\rangle} + \sqrt{0.81e^{i\theta_{23}} B_2\rangle})$ |
| $a_2$ | $(\sqrt{0.8e^{i\theta_{13}} A_2\rangle} + \sqrt{0.2e^{i\theta_{13}} A_3\rangle}) \otimes (\sqrt{0.43e^{i\theta_{13}} B_1\rangle} + \sqrt{0.57e^{i\theta_{13}} B_3\rangle})$   | $(\sqrt{0.14e^{i\theta_{12}} A_1\rangle} + \sqrt{0.86e^{i\theta_{12}} A_2\rangle}) \otimes (\sqrt{0.75e^{i\theta_{23}} B_2\rangle} + \sqrt{0.25e^{i\theta_{23}} B_3\rangle})$ |
| $a_3$ | $(\sqrt{0.9e^{i\theta_{12}} A_2\rangle} + \sqrt{0.1e^{i\theta_{12}} A_3\rangle}) \otimes (\sqrt{0.57e^{i\theta_{12}} B_2\rangle} + \sqrt{0.43e^{i\theta_{12}} B_3\rangle})$   | $(\sqrt{0.43e^{i\theta_{13}} A_2\rangle} + \sqrt{0.57e^{i\theta_{13}} A_3\rangle}) \otimes (\sqrt{0.34e^{i\theta_{23}} B_1\rangle} + \sqrt{0.66e^{i\theta_{23}} B_2\rangle})$ |
| $a_4$ | $(\sqrt{0.5e^{i\theta_{12}} A_2\rangle} + \sqrt{0.5e^{i\theta_{12}} A_3\rangle}) \otimes (\sqrt{0.5e^{i\theta_{12}} B_1\rangle} + \sqrt{0.5e^{i\theta_{12}} B_2\rangle})$     | $(\sqrt{0.9e^{i\theta_{12}} A_2\rangle} + \sqrt{0.1e^{i\theta_{12}} A_3\rangle}) \otimes (\sqrt{0.7e^{i\theta_{12}} B_3\rangle} + \sqrt{0.3e^{i\theta_{12}} B_1\rangle})$     |

**Table A.2**Initial evaluation information matrix for decision makers  $d_3$ .

|       | $c_1$                                                                                                                                                                         | $c_2$                                                                                                                                                                         |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $a_1$ | $(\sqrt{0.78e^{i\theta_{13}} A_2\rangle} + \sqrt{0.22e^{i\theta_{13}} A_3\rangle}) \otimes (\sqrt{0.61e^{i\theta_{12}} B_1\rangle} + \sqrt{0.39e^{i\theta_{12}} B_2\rangle})$ | $(\sqrt{0.28e^{i\theta_{12}} A_1\rangle} + \sqrt{0.72e^{i\theta_{12}} A_2\rangle}) \otimes (\sqrt{0.5e^{i\theta_{13}} B_1\rangle} + \sqrt{0.5e^{i\theta_{13}} B_3\rangle})$   |
| $a_2$ | $(\sqrt{0.4e^{i\theta_{13}} A_1\rangle} + \sqrt{0.6e^{i\theta_{13}} A_3\rangle}) \otimes (\sqrt{0.4e^{i\theta_{23}} B_1\rangle} + \sqrt{0.6e^{i\theta_{23}} B_3\rangle})$     | $(\sqrt{0.47e^{i\theta_{13}} A_1\rangle} + \sqrt{0.53e^{i\theta_{13}} A_3\rangle}) \otimes (\sqrt{0.26e^{i\theta_{23}} B_2\rangle} + \sqrt{0.74e^{i\theta_{23}} B_3\rangle})$ |
| $a_3$ | $(\sqrt{0.3e^{i\theta_{13}} A_1\rangle} + \sqrt{0.7e^{i\theta_{13}} A_3\rangle}) \otimes (\sqrt{0.75e^{i\theta_{13}} B_1\rangle} + \sqrt{0.25e^{i\theta_{13}} B_3\rangle})$   | $(\sqrt{0.36e^{i\theta_{23}} A_1\rangle} + \sqrt{0.64e^{i\theta_{23}} A_2\rangle}) \otimes (\sqrt{0.6e^{i\theta_{12}} B_1\rangle} + \sqrt{0.4e^{i\theta_{12}} B_3\rangle})$   |
| $a_4$ | $(\sqrt{0.39e^{i\theta_{12}} A_1\rangle} + \sqrt{0.61e^{i\theta_{12}} A_2\rangle}) \otimes (\sqrt{0.5e^{i\theta_{23}} B_1\rangle} + \sqrt{0.5e^{i\theta_{23}} B_2\rangle})$   | $(\sqrt{0.7e^{i\theta_{12}} A_1\rangle} + \sqrt{0.3e^{i\theta_{12}} A_2\rangle}) \otimes (\sqrt{0.48e^{i\theta_{12}} B_1\rangle} + \sqrt{0.52e^{i\theta_{12}} B_2\rangle})$   |
|       | $c_3$                                                                                                                                                                         | $c_4$                                                                                                                                                                         |
| $a_1$ | $(\sqrt{0.74e^{i\theta_{23}} A_2\rangle} + \sqrt{0.26e^{i\theta_{23}} A_3\rangle}) \otimes (\sqrt{0.31e^{i\theta_{12}} B_1\rangle} + \sqrt{0.69e^{i\theta_{12}} B_2\rangle})$ | $(\sqrt{0.47e^{i\theta_{12}} A_1\rangle} + \sqrt{0.53e^{i\theta_{12}} A_2\rangle}) \otimes (\sqrt{0.84e^{i\theta_{13}} B_1\rangle} + \sqrt{0.16e^{i\theta_{13}} B_3\rangle})$ |
| $a_2$ | $(\sqrt{0.37e^{i\theta_{23}} A_2\rangle} + \sqrt{0.63e^{i\theta_{23}} A_3\rangle}) \otimes (\sqrt{0.24e^{i\theta_{23}} B_1\rangle} + \sqrt{0.76e^{i\theta_{23}} B_2\rangle})$ | $(\sqrt{0.52e^{i\theta_{12}} A_1\rangle} + \sqrt{0.48e^{i\theta_{12}} A_2\rangle}) \otimes (\sqrt{0.37e^{i\theta_{23}} B_3\rangle} + \sqrt{0.63e^{i\theta_{23}} B_1\rangle})$ |
| $a_3$ | $(\sqrt{0.9e^{i\theta_{13}} A_2\rangle} + \sqrt{0.1e^{i\theta_{13}} A_3\rangle}) \otimes (\sqrt{0.24e^{i\theta_{12}} B_2\rangle} + \sqrt{0.76e^{i\theta_{12}} B_3\rangle})$   | $(\sqrt{0.43e^{i\theta_{12}} A_2\rangle} + \sqrt{0.57e^{i\theta_{12}} A_3\rangle}) \otimes (\sqrt{0.75e^{i\theta_{12}} B_1\rangle} + \sqrt{0.25e^{i\theta_{12}} B_3\rangle})$ |
| $a_4$ | $(\sqrt{0.46e^{i\theta_{23}} A_2\rangle} + \sqrt{0.54e^{i\theta_{23}} A_3\rangle}) \otimes (\sqrt{0.28e^{i\theta_{12}} B_1\rangle} + \sqrt{0.72e^{i\theta_{12}} B_2\rangle})$ | $(\sqrt{0.34e^{i\theta_{13}} A_1\rangle} + \sqrt{0.66e^{i\theta_{13}} A_3\rangle}) \otimes (\sqrt{0.75e^{i\theta_{12}} B_2\rangle} + \sqrt{0.25e^{i\theta_{12}} B_3\rangle})$ |

**Data availability**

No data was used for the research described in the article.

**Table A.3**Initial evaluation information matrix for decision makers  $d_4$ .

|       | $c_1$                                                                                                                                                                         | $c_2$                                                                                                                                                                         |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $a_1$ | $(\sqrt{0.14e^{i\theta_{12}} A_1\rangle} + \sqrt{0.86e^{i\theta_{12}} A_3\rangle}) \otimes (\sqrt{0.27e^{i\theta_{12}} B_1\rangle} + \sqrt{0.73e^{i\theta_{12}} B_2\rangle})$ | $(\sqrt{0.33e^{i\theta_{12}} A_1\rangle} + \sqrt{0.67e^{i\theta_{12}} A_3\rangle}) \otimes (\sqrt{0.42e^{i\theta_{23}} B_2\rangle} + \sqrt{0.58e^{i\theta_{23}} B_3\rangle})$ |
| $a_2$ | $(\sqrt{0.47e^{i\theta_{13}} A_1\rangle} + \sqrt{0.53e^{i\theta_{13}} A_3\rangle}) \otimes (\sqrt{0.47e^{i\theta_{23}} B_2\rangle} + \sqrt{0.53e^{i\theta_{23}} B_3\rangle})$ | $(\sqrt{0.7e^{i\theta_{13}} A_1\rangle} + \sqrt{0.3e^{i\theta_{13}} A_3\rangle}) \otimes (\sqrt{0.57e^{i\theta_{23}} B_1\rangle} + \sqrt{0.43e^{i\theta_{23}} B_3\rangle})$   |
| $a_3$ | $(\sqrt{0.9e^{i\theta_{23}} A_2\rangle} + \sqrt{0.1e^{i\theta_{23}} A_3\rangle}) \otimes (\sqrt{0.8e^{i\theta_{12}} B_1\rangle} + \sqrt{0.2e^{i\theta_{12}} B_2\rangle})$     | $(\sqrt{0.36e^{i\theta_{23}} A_1\rangle} + \sqrt{0.64e^{i\theta_{23}} A_2\rangle}) \otimes (\sqrt{0.6e^{i\theta_{12}} B_1\rangle} + \sqrt{0.4e^{i\theta_{12}} B_3\rangle})$   |
| $a_4$ | $(\sqrt{0.52e^{i\theta_{12}} A_1\rangle} + \sqrt{0.48e^{i\theta_{12}} A_2\rangle}) \otimes (\sqrt{0.43e^{i\theta_{12}} B_1\rangle} + \sqrt{0.57e^{i\theta_{12}} B_2\rangle})$ | $(\sqrt{0.7e^{i\theta_{13}} A_1\rangle} + \sqrt{0.3e^{i\theta_{13}} A_2\rangle}) \otimes (\sqrt{0.57e^{i\theta_{13}} B_1\rangle} + \sqrt{0.43e^{i\theta_{13}} B_3\rangle})$   |
|       | $c_3$                                                                                                                                                                         | $c_4$                                                                                                                                                                         |
| $a_1$ | $(\sqrt{0.47e^{i\theta_{23}} A_2\rangle} + \sqrt{0.53e^{i\theta_{23}} A_3\rangle}) \otimes (\sqrt{0.59e^{i\theta_{12}} B_1\rangle} + \sqrt{0.41e^{i\theta_{12}} B_2\rangle})$ | $(\sqrt{0.47e^{i\theta_{12}} A_1\rangle} + \sqrt{0.53e^{i\theta_{12}} A_3\rangle}) \otimes (\sqrt{0.85e^{i\theta_{23}} B_1\rangle} + \sqrt{0.15e^{i\theta_{23}} B_2\rangle})$ |
| $a_2$ | $(\sqrt{0.46e^{i\theta_{12}} A_1\rangle} + \sqrt{0.54e^{i\theta_{12}} A_3\rangle}) \otimes (\sqrt{0.83e^{i\theta_{23}} B_2\rangle} + \sqrt{0.17e^{i\theta_{23}} B_3\rangle})$ | $(\sqrt{0.9e^{i\theta_{13}} A_2\rangle} + \sqrt{0.1e^{i\theta_{13}} A_3\rangle}) \otimes (\sqrt{0.6e^{i\theta_{23}} B_1\rangle} + \sqrt{0.4e^{i\theta_{23}} B_3\rangle})$     |
| $a_3$ | $(\sqrt{0.9e^{i\theta_{23}} A_2\rangle} + \sqrt{0.1e^{i\theta_{23}} A_3\rangle}) \otimes (\sqrt{0.71e^{i\theta_{13}} B_1\rangle} + \sqrt{0.29e^{i\theta_{13}} B_2\rangle})$   | $(\sqrt{0.25e^{i\theta_{12}} A_1\rangle} + \sqrt{0.75e^{i\theta_{12}} A_3\rangle}) \otimes (\sqrt{0.26e^{i\theta_{23}} B_1\rangle} + \sqrt{0.74e^{i\theta_{23}} B_2\rangle})$ |
| $a_4$ | $(\sqrt{0.96e^{i\theta_{12}} A_1\rangle} + \sqrt{0.04e^{i\theta_{12}} A_2\rangle}) \otimes (\sqrt{0.96e^{i\theta_{12}} B_1\rangle} + \sqrt{0.04e^{i\theta_{12}} B_2\rangle})$ | $(\sqrt{0.47e^{i\theta_{12}} A_1\rangle} + \sqrt{0.53e^{i\theta_{12}} A_3\rangle}) \otimes (\sqrt{0.5e^{i\theta_{12}} B_1\rangle} + \sqrt{0.5e^{i\theta_{12}} B_2\rangle})$   |

**Table A.4**Trust path weight  $w_r$  ( $r = 1, 2, \dots, 9$ ).

| Trust path                                            | Weight | Trust path                                            | Weight |
|-------------------------------------------------------|--------|-------------------------------------------------------|--------|
| $d_1 \rightarrow d_2 \rightarrow d_3$                 | 0.4    | $d_3 \rightarrow d_1 \rightarrow d_2$                 | 0.3    |
| $d_1 \rightarrow d_4 \rightarrow d_3$                 | 0.6    | $d_3 \rightarrow d_4 \rightarrow d_2$                 | 0.3    |
| $d_2 \rightarrow d_3 \rightarrow d_1$                 | 0.6    | $d_3 \rightarrow d_4 \rightarrow d_1 \rightarrow d_2$ | 0.4    |
| $d_2 \rightarrow d_3 \rightarrow d_2$                 | 0.4    | $d_4 \rightarrow d_2 \rightarrow d_3$                 | 0.4    |
| $d_4 \rightarrow d_1 \rightarrow d_2 \rightarrow d_3$ | 0.6    |                                                       |        |

**Table A.5**Bounded confidence levels  $B^k$  ( $k = 1, 2, \dots, 4$ ) for decision-makers.

|       | $c_1$ | $c_2$ | $c_3$ | $c_4$ |
|-------|-------|-------|-------|-------|
| $d_1$ | 0.15  | 0.1   | 0.2   | 0.3   |
| $d_2$ | 0.05  | 0.15  | 0.1   | 0.16  |
| $d_3$ | 0.3   | 0.45  | 0.21  | 0.04  |
| $d_4$ | 0.2   | 0.15  | 0.2   | 0.25  |

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